NATIONAL CLINICAL GUIDELINES

The Diagnosis & Management of Diabetes in Pregnancy

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المبادئ الإرشادية السريرية لدولة قطر
NATIONAL CLINICAL GUIDELINES FOR QATAR

وزارة الصحة العامة
Ministry of Public Health
State of Qatar • دولة قطر

Version History

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1.0	Final	24th April 2017	Guidelines Team
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3.0	Final	24th April 2024	Guidelines Team

Citation

Suggested citation style:

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Abbreviations

The abbreviations used in these guidelines are as follows:

ACE	Angiotensin Converting Enzyme
ACR	Albumin-Creatinine Ratio
AFI	Amniotic Fluid Index
ARBs	Angiotensin Receptor Blockers
BG	Blood Glucose
BMI	Body Mass Index
CGMS	Continuous Glucose Monitoring System
CKD	Chronic Kidney Disease
CTG	Cardiotocography
DKA	Diabetic Ketoacidosis
DM	Diabetes Mellitus
dwAG	Doi's weighted average glucose
eGFR	Estimated Glomerular Filtration Rate
FBG, 1hBG, 2hBG	Fasting Blood Glucose (Venous), 1h and 2h BG on the OGTT
GDG	Guideline Development Group
GDM	Gestational Diabetes Mellitus
GTT	Glucose Tolerance Test
HbA_{1C}	Glycated Haemoglobin
hGDM	High-risk Gestational Diabetes Mellitus
HHS	Hyperglycaemic Hyperosmolar State
IADPSG	International Association of Diabetes and Pregnancy Study Groups

IDF	International Diabetes Federation
IGG	Impaired Gestational Glycaemia
IGT	Impaired Glucose Tolerance
IOL	Induction of Labour
IUFD	Intrauterine Fetal Death
IUGR	Intrauterine Growth Restriction
MDI	Multi-Dose Injection
MiTy	Multi-center randomized controlled trial
NCGPC	National Clinical Guidelines & Pathways Committee
NGG	Normal gestational glycaemia
NICU	Neonatal Intensive Care Unit
NPH	Neutral Protamine Hagedorn
NPRP	National Priorities Research Program
OGTT	Oral Glucose Tolerance Test
PCOS	Polycystic Ovary Syndrome
RAAS	Renin-Angiotensin-Aldosterone System
RCTs	Randomized Controlled Trials
RDS	Respiratory Distress Syndrome
SGA	Small for Gestational Age
SMBG	Self-Monitoring of Blood Glucose
T1DM	Type 1 Diabetes Mellitus
T2DM	Type 2 Diabetes Mellitus
TAR	Time above range
TBR	Time below range
TIR	Time in range
TOBOGM	The Treatment of Booking Gestational Diabetes Mellitus
TSH	Thyroid Stimulating Hormone
VRII	Variable Rate Insulin Infusion
WHO	World Health Organization

Table of Contents

1 Information About These Guidelines

1.1 Objective and Purpose of the Guidelines

The purpose of these guidelines is to define the appropriate diagnosis and management of diabetes mellitus in pregnancy. The objective is to guide the appropriate investigation, prescribing and referral of patients presenting to provider organisations in Qatar.

It is intended that the guidelines will be used primarily by midwives, nurses, health educators/counsellors, dieticians, and all physicians responsible for women with gestational diabetes or those with diabetes who become, or are intending to become, pregnant.

1.2 Scope of the Guidelines

Aspects of care covered in these guidelines include the following:

- Assessment and management of diabetes mellitus in pregnancy including:
    - Pre-conception counselling care.
    - Screening for diabetes in pregnancy.
    - Management of gestational diabetes and women with pre-existing diabetes mellitus.
    - Intrapartum and postnatal considerations.
    - Management of common complications of Diabetes in pregnancy.

1.3 Editorial Approach

This guidelines document has been developed and issued by the Ministry of Public Health of Qatar (MoPH), through a process which aligns with international best practices in guidelines development and localisation. The guidelines will be reviewed and updated to ensure alignment with the latest evidence and to incorporate comments and feedback from stakeholders across Qatar.

The editorial methodology, used to develop these guidelines, has involved the following critical steps:

- Extensive literature search for well-reputed published evidence relating to the topic.
- Critical appraisal of the literature.
- Development of a draft summary guidelines.
- Review of the summary guidelines with Guideline Development Group, comprised of practising healthcare professionals, subject matter experts, evidence-based medicine experts and patient representatives, from across Qatar.
- Independent review of the guidelines by the National Clinical Guidelines & Pathways Committee, appointed by the MoPH, from amongst stakeholder organisations across Qatar.

Whilst the MoPH has facilitated the development of the guidelines, the MoPH has not influenced the specific recommendations made within it.

1.4 Sources of Evidence

The scientific literature has been systematically queried using specially developed, customised, and tested search strings. Search strategies are developed to allow an efficient yet comprehensive analysis of relevant publications for a given topic and to maximise retrieval of articles with certain desired characteristics pertinent to a guideline.

All retrieved publications have been individually reviewed and assessed in terms of quality, utility, and relevance. Preference is given to publications that:

1. Are designed with rigorous scientific methodology.
2. Are published in higher-quality journals.
3. Address an aspect of specific importance to the guidelines in question.

Further information about the literature search and appraisal process is included in the appendix.

1.5 Evidence Levelling and Grading of Recommendations

Recommendations made within these guidelines are supported by evidence from the medical literature and where possible, the most authoritative sources have been used in the development of these guidelines.

To achieve transparency and simplicity of the recommendations, the GRADE system is implemented with slight modification where the low and very low categories are combined. The following evidence hierarchy has been used to grade the level of authoritativeness of the evidence used, where recommendations have been made within this guideline [1].

- **Level 1 (L1):**
  - Meta-analyses of randomized controlled trials.
  - Randomized controlled trials.
- **Level 2 (L2):**
  - Meta-analyses of observational studies.
  - Observational studies, examples include:
    - Cohort studies with statistical adjustment for potential confounders.
    - Cohort studies without adjustment.
    - Case series with historical or literature controls.
    - Uncontrolled case series.
- **Level 3 (L3):**
  - Expert opinion.
  - Case reports.
  - Statements in published articles or textbooks.
  - Unpublished data, examples include:
    - Large database analyses.
    - Written protocols or outcomes reports from large practices.

To give additional insight into the reasoning underlying certain recommendations and the strength of recommendations, the following recommendation grading has been used, where recommendations are made:

- **Recommendation Grade A (RGA):** Evidence demonstrates at least moderate certainty of a net benefit from the recommendation.
- **Recommendation Grade B (RGB):** Evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended.
- **Recommendation Grade C (RGC):** Evidence demonstrates potential harm that outweighs benefit; additional research is recommended.
- **Recommendation of the GDG (R-GDG):** Recommended based on the clinical experience of the Guideline Development Group members.

1.6 Guideline Development Group Members

The following table lists members of the Guideline Development Group (GDG) nominated by their respective organisations and the National Clinical Guidelines & Pathways Committee. The GDG members have reviewed and provided their feedback and approval of these guidelines document. Each member has completed a declaration of conflicts of interest, which has been reviewed and retained by the MoPH.

Guideline Development Group Members		
Name (ordered as per surnames)	**Title**	**Organisation**
Prof Abdul Badi Abou-Samra	Director, Qatar Metabolic Institute, Chief Quality Officer, Deputy, Chief Medical Officer, Professor of Medicine	Hamad Medical Corporation & Weill Cornell Medicine - Qatar
Hayam Mosed Al-Hajaji	Clinical Dietitian	Hamad Medical Corporation
Dr Alia Aleidi AlRuwaili	Senior Consultant Family Medicine	Primary Health Care Corporation
Naglaa Alsharkawy	Senior Diabetes Nursing Educator	Hamad Medical Corporation
Moufida Bent Belgacem Azek	Midwife -Women Wellness & Research Center	Hamad Medical Corporation
Dr Ahmed M. Hussein Babiker[1]	Head of Registration Section & Clinical Pharmacist	Dept of Pharmacy and Drug Control, MOPH
Dr Mohammed Bashir	Senior Consultant Endocrinologist	Hamad Medical Corporation
Dr Stephen Beer	Senior Consultant Endocrinologist/ Diabetologist	Hamad Medical Corporation
Prof Suhail A. R. Doi[2]	Head, Department of Population Medicine, College of Medicine	Qatar University
Dr Ibrahim Mamoun Ibrahim	Senior Attending Physician, Medicine/ Endocrinologist, Assistant Professor of clinical Medicine, Weill Cornell Medicine- Qatar	Sidra Medicine
Dr Gbemisola Okunoye	Senior Attending Physician – Obstetrics, Assistant Professor of Obstetrics & Gynaecology	Sidra Medicine
Dr Hessa Ibrahim Shahbic	Senior Consultant in Community Medicine, Assistant Director of Medicine for Women Health, Clinical Affairs Department	Primary Health Care Corp
Dr Faten Al Taher Mohd Taha	Senior Consultant OB-GYN, Women's Wellness and Research Center	Hamad Medical Corporation
Dr Mahmoud Ali Zirie	Senior Consultant & Division Chief of Endocrine Division	Hamad Medical Corporation

[1] Dr Ahmed Babiker attended the MOPH in his capacity as a Clinical Pharmacist and advisor on the availability of medications in Qatar.

[2] This GDG member was funded by Program Grant #NPRP 10-0129-170274 from the Qatar National Research Fund.

1.7 National Clinical Guidelines & Pathways Committee Members

The following table lists members of the National Clinical Guidelines & Pathways Committee (NCGPC), appointed by the MoPH. The NCGPC members have reviewed and provided their feedback and approval of the guideline document. Each member has completed a declaration of conflicts of interest, which has been reviewed and retained by the MoPH.

National Clinical Guidelines & Pathways Committee (NCGPC) Members		
Name	**Title**	**Organisation**
Ms Huda Amer Al-Katheeri	Chair of the NCGPC, Director of Strategic Planning and Performance and Innovation Department	Ministry of Public Health
Shk Dr Mohammed Hamad J. Al Thani	Co-Chair of the NCGPC, Non-Communicable Diseases Prevention Programs Director	Ministry of Public Health
Dr Hani Ben Hassen Al Kilani	Senior Consultant, Executive Director for Corporate Clinical Policy and Guidelines	Hamad Medical Corporation
Dr Ibtihal Abdelgadir	Lead Clinical Practice Guidelines Committee	Sidra Medicine
Dr Alshaymaa Mohammed A. M. Al-Motawa	Consultant Family Medicine, Lead Medical Officer (Doha)	QatarEnergy
Dr Basil Bashqawi	Supervisor, Accreditation & Healthcare Professional Education, Health Professions Department	Ministry of Public Health
Dr Abi Khalil Charbel	Professor of Medicine and Genetic Medicine, Assistant Dean of International Affairs	Weill Cornell Medicine-Qatar
Dr Paul Dijkstra	Director of Medical Education	ASPETAR
Dr Mohammed Elrishi	Senior Consultant Endocrine, Diabetes, and Internal Medicine	Al Ahli Hospital
Dr Dahlia Mustafa Hassan	Senior Consultant Family Medicine	Primary Health Care Corporation
Dr Ghassan Youseph Hommos	Senior Consultant Endocrinology	Al Emadi Hospital
Dr Marwan Abu-Hijleh	Professor and Dean	College of Medicine, Qatar University

1.8 Responsibilities of Healthcare Professionals

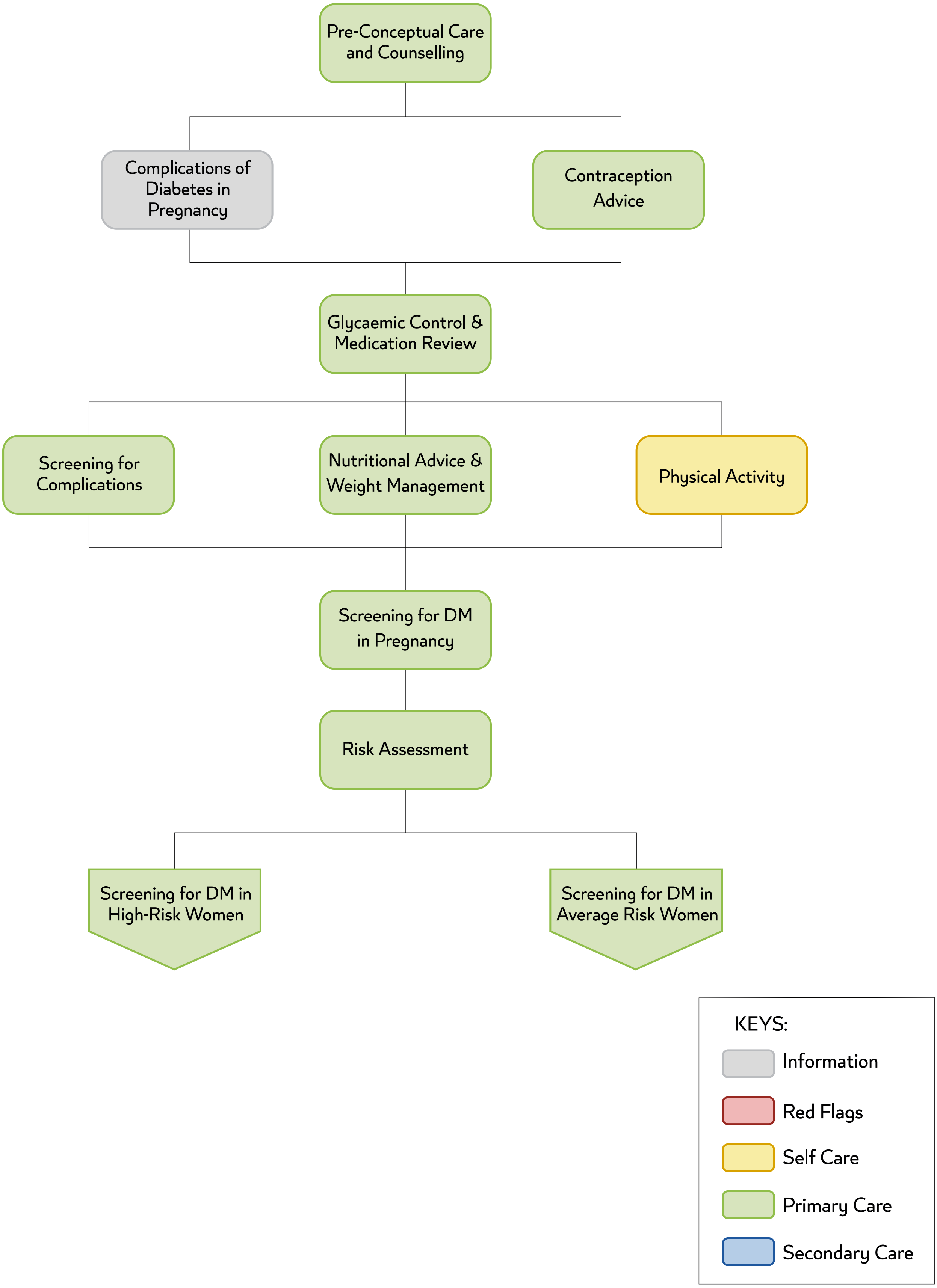
These guidelines have been issued by the MOPH to define how care should be provided in Qatar. It is based upon a comprehensive assessment of the evidence as well as its applicability to the national context of Qatar. Healthcare professionals are expected to take these guidelines into account when exercising their clinical judgement in the care of patients presenting to them.

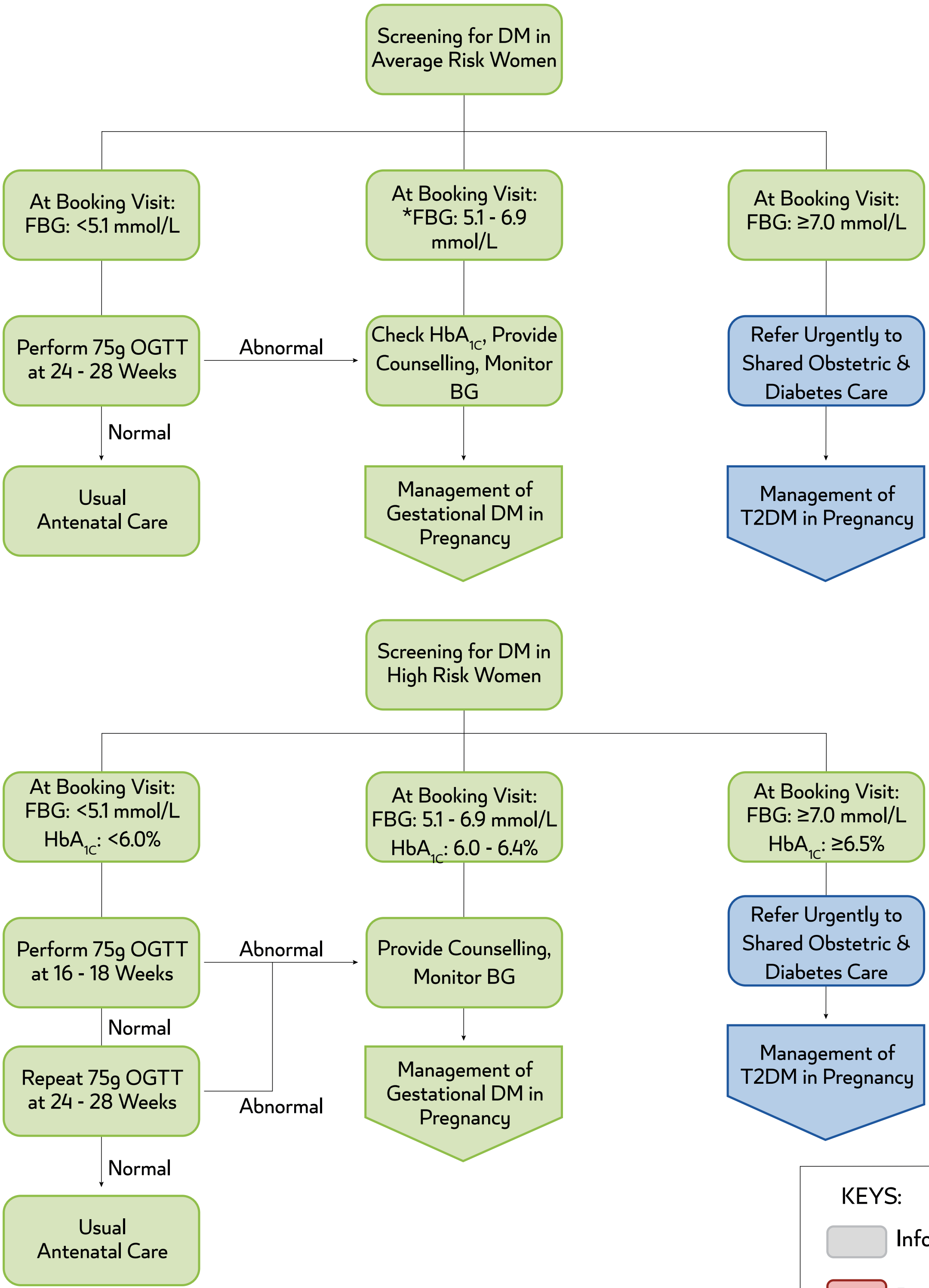
This guideline does not override individual professional responsibility to take decisions which are appropriate to the circumstances of the patient concerned. Such decisions should be made in consultation with the patient, their guardians, or caregivers and should consider the individual risks and benefits of any intervention that is contemplated in the patient's care.

2 Diabetes in Pregnancy Pathway

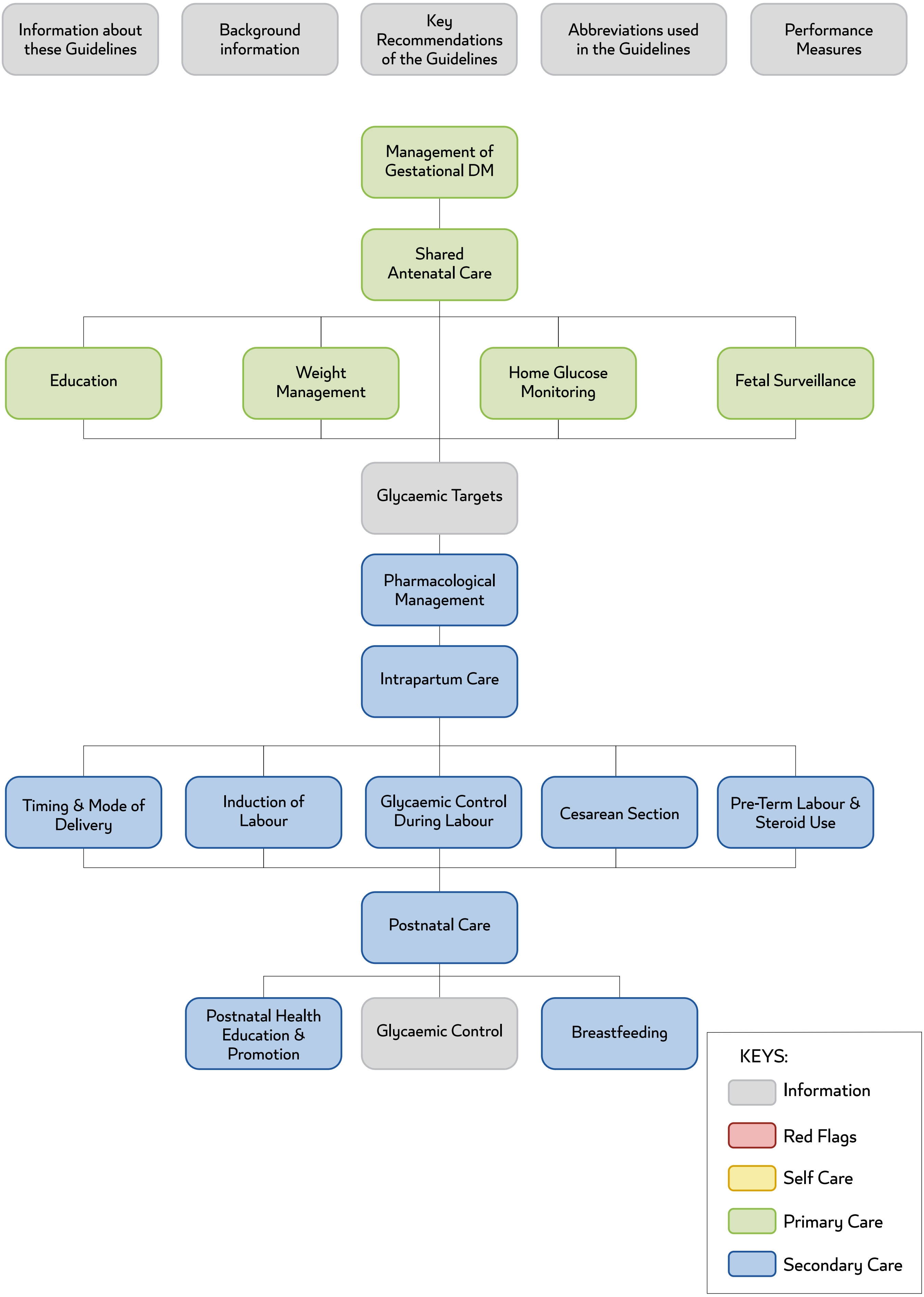
Click on a box below to see the relevant page of the Pathway.

- Information about these Guidelines
- Background information
- Key Recommendations of the Guidelines
- Abbreviations used in the Guidelines
- Performance Measures





*If FBG is 5.1-5.3, consider following OGTT at 16-18 weeks [R-GDG].


Information about these Guidelines
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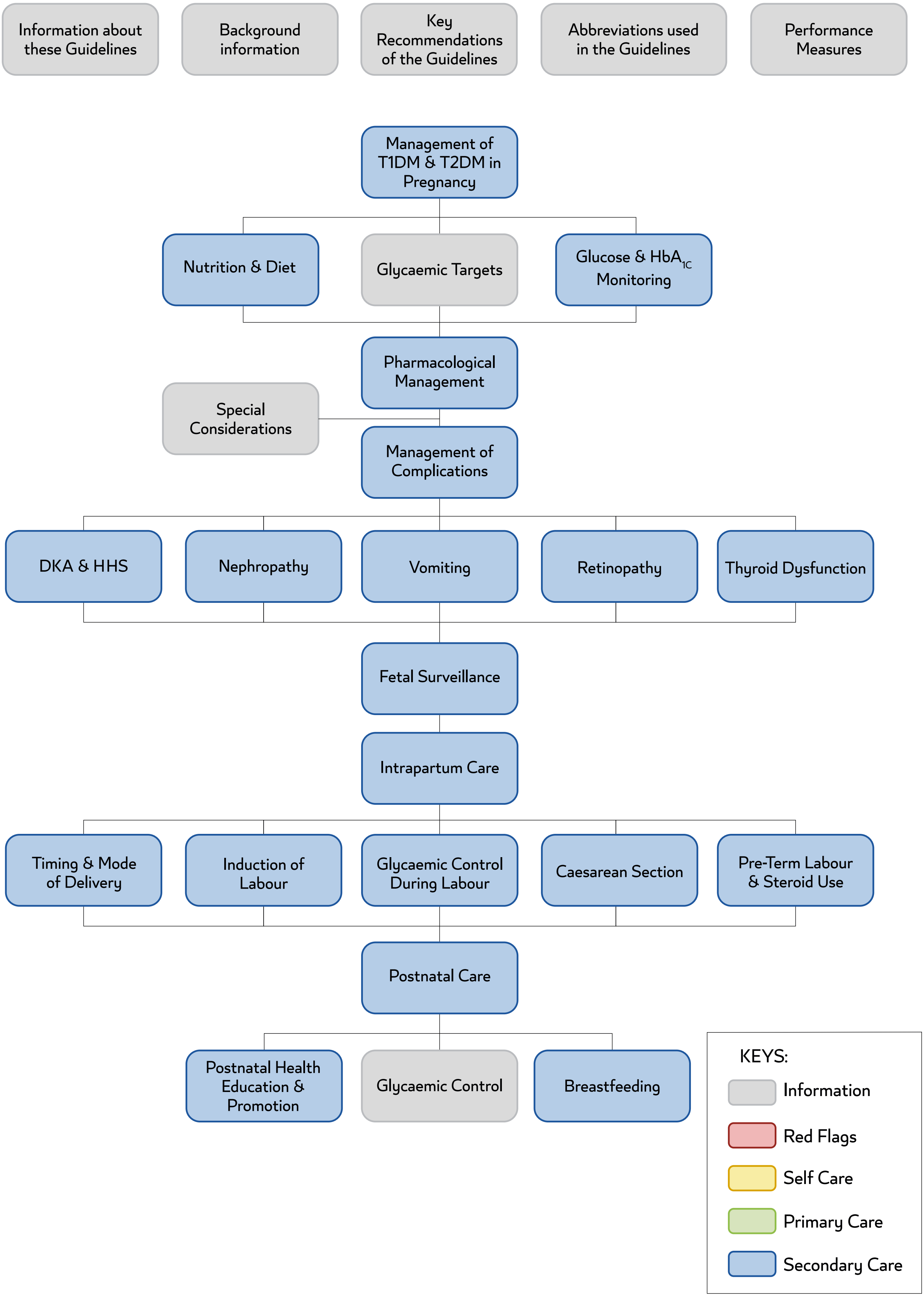
Abbreviations used in the Guidelines
Performance Measures
Management of Gestational DM
Shared Antenatal Care
Education
Weight Management
Home Glucose Monitoring
Fetal Surveillance
Glycaemic Targets
Pharmacological Management
Intrapartum Care
Timing & Mode of Delivery
Induction of Labour
Glycaemic Control During Labour
Cesarean Section
Pre-Term Labour & Steroid Use
Postnatal Care
Postnatal Health Education & Promotion
Glycaemic Control
Breastfeeding
KEYS:
Information
Red Flags
Self Care
Primary Care
Secondary Care


Information about these Guidelines
Background information
Key Recommendations of the Guidelines
Abbreviations used in the Guidelines
Performance Measures
Management of T1DM & T2DM in Pregnancy
Nutrition & Diet
Glycaemic Targets
Glucose & HbA1C Monitoring
Pharmacological Management
Special Considerations
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Vomiting
Retinopathy
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Fetal Surveillance
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Timing & Mode of Delivery
Induction of Labour
Glycaemic Control During Labour
Caesarean Section
Pre-Term Labour & Steroid Use
Postnatal Care
Postnatal Health Education & Promotion
Glycaemic Control
Breastfeeding
KEYS:
Information
Red Flags
Self Care
Primary Care
Secondary Care

3 Key Recommendations of the Guidelines

The key recommendations of these guidelines are:

Pre-Conception Care of Women with Pre-existing Diabetes who may become pregnant *(Section 5)*:

- Women with known diabetes who may become pregnant should receive pre-conception care [2] **[L2, RGA]**.
- Pre-pregnancy care should be part of routine diabetes care irrespective of the setting **[R-GDG]**.
- Care in a specialist pre-conception clinic should be provided jointly by the adult diabetes services and the maternity service for women wishing to become pregnant **[R-GDG]**.
- Aim to keep HbA_{1C} <6.5%, prior to conception, if it can be achieved without problematic hypoglycaemia[2,3] **[L2, RGA]**.
- If HbA_{1C} is ≥10%, the absolute risk of congenital malformation increases significantly, and women should therefore be advised not to become pregnant[3]. The risk of malformations may also be increased at lower HbA_{1C} levels **[R-GDG]**.
- Women with T2DM intending pregnancy should be managed with[2,3]:
    - Lifestyle modification (diet and increased physical activity). This should be integrated in the care of every patient with diabetes, if not sufficient medication can be added.
    - Metformin has been shown to be safe and efficacious during pregnancy[4-6].
    - The safety profile of other oral hypoglycaemic agents is not known and should be stopped prior to conception[7-10].
- Insulin use in pregnancy[3,7-11]:
    - Regular human insulin and intermediate acting insulin, NPH-insulin, used to be the main insulin preparations for diabetes in pregnancy; however, they may not be convenient to the patient because of kinetics and duration of action.
    - The rapid-acting insulin analogues – aspart and lispro, the ultra-fast insulin – fast- aspart. and the longer acting analogue, insulin detemir, and degludec are approved in pregnancy. They are more advantageous than soluble human insulin during pregnancy in reducing the risk of hypoglycaemia.
    - Observational studies did not show differences in pregnancy and fetal outcome between, insulin glargine and other insulins. If a woman with diabetes is in good control with these analogues, they can be continued during pregnancy; however, it is not recommended to start them during pregnancy.
    - Other insulin analogues have not been studied in pregnancy and therefore have not been approved for this indication.

Screening for Diabetes in Pregnancy *(Section 6)***:**

- See *Sections 6.1-6.2* for screening tests to be undertaken in average-risk and high-risk women during pregnancy.
- See *Section 6.3* for referral criteria for women who are diagnosed with GDM or diabetes during pregnancy.

Management of Gestational Diabetes Mellitus (GDM) *(Section 7)***:**

- Antenatal care for women with GDM should be provided by a multidisciplinary team **[R-GDG].**
- Change in lifestyle is essential in the management of women with GDM, and in some cases may be the only treatment required[2,3] **[L1]**:
    - Women should adhere to the acceptable weight gain limits in *Section 7.3* during pregnancy in order to minimise the risk of complications, but active weight loss should not be attempted **[R-GDG]**.
    - All patients with GDM should be referred to a dietitian for dietary planning[2,3] **[L2]**.
- Every woman should have a reliable and recently calibrated glucometer at home[3].
- Women on diet management alone, Metformin therapy or single dose of basal insulin, should be advised to monitor their BG levels, four times per day[3].
- Patients on multi-dose injections of insulin should be advised to monitor their BG levels, up to 6-7 times per day[3,12].

- If achievable without causing problematic hypoglycaemia, control should aim for[3,11-13] **[L1, RGA]**:
  - Fasting BG ≤5.3 mmol/L (<95 mg/dL).
  - One hour after meals ≤7.8 mmol/L (<140mg/dL).
  - Two hours after meals ≤6.7 mmol/L (<120 mg/dL).
- Continuous glucose monitoring could be offered to women with GDM treated with insulin **[R-GDG]**
  - The target glucose should be set between 3.5-7.8 mmol/L (63-140 mg/dl).
  - Time above Range (TAR) should be ≤ 15%.
  - Time in Range (TIR) should be ≥ 80%.
  - Time below Range (TBR) should be < 4%.
- Regular monitoring of HbA_{1C} is not recommended after measurement at initial diagnosis[3] **[L2]**.
- Metformin should be offered to patients with GDM if BG targets are not being achieved using diet.
- and exercise within 1-2 weeks[3,14] **[L1, RGA]** *(see Section 7.7.1)*.
- Insulin should be started if glycaemic control cannot be achieved with Metformin; when Metformin is contraindicated or not acceptable to patients[3] *(see Section 7.7.2)*.

Caution:

- Metformin should not be used in women with CKD (eGFR < 60 ml/min/1.72m^3).
- Consider avoidance of Metformin in women with sonographic evidence of small-for gestational age (SGA) baby defined as abdominal cicumference <10%.

Fetal Surveillance in Women with GDM *(Section 7.6)***:**

- Fetal surveillance includes[3]:
  - A viability scan and confirmation of gestational age, preferably at 7-9 weeks, but no later than 12 weeks, where possible.
  - Nuchal translucency scan at 11-13 weeks.
  - Fetal anomaly scan at 18-20 weeks.
  - Fetal growth and amniotic fluid volume assessment at 28-32 weeks and again at 36 weeks.
  - Further scans should depend upon glycaemic control and clinical judgment.
  - Maternal surveillance of fetal movements from 24 weeks onwards **[R-GDG]**.

Management of Women with T1DM or T2DM in Pregnancy *(Section 8)***:**

- Every patient should have a reliable and recently calibrated glucometer at home[3].
- Patients on diet management alone, Metformin therapy or single dose of basal insulin, should be advised to monitor their BG levels, 4 times per day[3].
- Patient on multi-dose injections of insulin should be advised to monitor their BG levels, up to 6-7 times per day[3,11].
- If achievable without causing problematic hypoglycaemia, control should aim for [3,11,13,15] **[L1, RGA]**:
  - Fasting BG ≤5.3 mmol/L (<95 mg/dL).
  - One hour after meals ≤7.8 mmol/L (140mg/dL).
  - Two hours after meals ≤6.7 mmol/L (<120 mg/dL).
- Continuous glucose monitoring could be offered to all women with type 1 DM insulin[16]:
  - The target glucose should be set between 63-140 mg/dl (3.5-7.8 mmol/l).
  - Time above Range (TAR) should be ≤ 25%.
  - Time in Range (TIR) should be ≥ 70%.
  - Time below Range (TBR) should be < 4%.
- Continuous glucose monitoring could be offered to women with DM-Type 2 treated with insulin **[R-GDG]**:
  - The target glucose should be set between 63-140 mg/dl (3.5-7.8 mmol/l)

View Pathway

    - Time above Range (TAR) should be ≤ 15%.
    - Time in Range (TIR) should be ≥ 80%.
    - Time below Range (TBR) should be < 4%.
- HbA_{1C} should be measured in the first clinic review and then at least once in each trimester or more frequently, e.g. monthly [2,3,11].
- The HbA_{1C} target is ≤6.5%[2].
- A Multi-Dose Injection (MDI) regimen is recommended for the majority of patients[3,8,17].
- Patients with T2DM taking Metformin can be continued on their treatment as usual[3] **[L2]**.
- Patients should be empowered and taught how to adjust their own insulin doses.
- Aspirin 100 mg daily is recommended from 13 weeks onward to all women with pre-existing T1DM or T2DM[18] **[L1, RGA]**.

Caution:

- Avoid Metformin in women with uncontrolled hypertension, or nephropathy defined as ACR (>3.5 mg/mmol). The MiTy study reported a high rate of SGA in this subgroup[19].
- Avoid Metformin in women with CKD, defined as eGFR <60 ml/min/1.73m[3].

Fetal Surveillance in Women with T1DM or T2DM in Pregnancy *(Section 8.6)***:**

- Fetal surveillance includes[3]:
    - A viability scan and confirmation of gestational age, preferably at 7-9 weeks, but no later than 12 weeks, where possible.
    - Nuchal translucency scan at 11-13 weeks.
    - Fetal anomaly scan at 18-20 weeks.
    - Third trimester ultrasound assessment for[3]:
        - Fetal growth.
        - Assessment of liquor volume and fetal movements.
        - Estimation of fetal growth should be performed on a 4-weekly basis from 24 weeks, in order to assess fetal growth velocity.
        - More frequent ultrasound and fetal assessment will be determined by glycaemic control, fetal status or maternal health.
    - Advise delivery by Induction of Labour (IOL) or elective caesarean section (if indicated)[3]:
        - At 38-39 weeks, if there are no metabolic or other maternal or fetal complications and the diabetes is well controlled **[R-GDG]**.
        - Before 38 weeks, if metabolic or other maternal or fetal complications are present (including poor maternal glycaemic control).
- For women with GDM:
    - Advise delivery no later than 40 weeks plus 6 days, if glycaemic control is satisfactory[3].
        - Offer elective delivery by IOL or caesarean section if delivery has not occurred by then.
    - If glycaemic control is unsatisfactory plan for delivery, as for women with T1DM or T2DM in pregnancy, described above.

Glycaemic Control During Labour *(Section 10.2)***:**

- Hourly BG monitoring is recommended, aiming to keep BG between 4-7 mmol/L[3] **[L2]**.
- For patients using insulin[3]:
    - Should be allowed a light diet if desired.

  - Variable Rate Insulin Infusion (VRII), should be considered (as shown in Table 10.2) in the following situations:
    - Patients with T1DM.
    - All other patients with diabetes in pregnancy if the capillary BG is ≥ 7 mmol/L for two consecutive hours.
- For patients who are diet-controlled or using Metformin[3]:
  - VRII should only be commenced if the BG is ≥7 mmol/L for two consecutive hours.
- See *Sections 10.3 and 10.4* for recommendations in women undergoing induction of labour or elective caesarean section.

Women in Pre-Term Labour *(Section 10.5)***:**
- Diabetes is not a contraindication for steroid use[3].
- In the absence of contraindications, steroids should be given to women with diabetes in pregnancy in pre-term labour up to 36 weeks gestation[3].

Postnatal Care and Advice *(Section 11)***:**
- See recommendations in *Section 11.*

4 Background Information

4.1 Definition

Gestational Diabetes Mellitus (GDM) is defined as carbohydrate intolerance of variable severity, with onset or first recognition during pregnancy, that is neither pre-existing type 1 or type 2 diabetes[2].

Diabetes may predate conception and be either known to the patient or discovered during pregnancy[2,13,20]. Women who are discovered to have diabetes in the first trimester of pregnancy are classified as having diabetes in pregnancy, rather than GDM[2,13,20].

4.2 Epidemiology

The worldwide prevalence of hyperglycaemia in pregnancy is rising with the increasing prevalence of diabetes globally[20]. In 2021, the International Diabetes Federation (IDF) estimated that 16.7% of all live births in women aged 20-49, were affected by hyperglycaemia[20].

Analysis of GDM screening data from years 2019 through 2022 indicated that the prevalence of GDM in Qatar is in the range of 30%*, this is much higher than other countries such as Canada (17.8%) and France (12.1%)[21,22] A Qatar-based study demonstrated that 45% of GDM cases were seen in women aged between the age of 35-45 years[21,23].

It is expected that over the next 20 years, the regions of Middle East and North Africa will see the largest increase in the population of female diabetes patients[24]. 60% of women diagnosed with GDM subsequently develop type 2 diabetes (T2DM) within 4 years after delivery, increasing to 70% after 10 years[24].

* Analysis of GDM screening data from 2019, 2020, 2021 and 2022 showed a prevalence of GDM in the range of 28%-36%

5 Pre-Conception Counseling and Care

5.1 Pre-conception care

All women who intend to become pregnant should receive pre-conception care[2] **[L2, RGA]**.

The pre-pregnancy care outlined below should be part of the routine care of all women with pre-existing diabetes, irrespective of the setting. Women with diabetes who cannot be managed in a primary care setting should be referred to a specialist pre-conception clinic **[R-GDG]**.

Care in a specialist pre-conception clinic should be provided jointly by the adult diabetes services and the maternity services for women wishing to become pregnant **[R-GDG]**.

The pre-conception care includes the following [2,13]:

- Health education and counselling on the risk of diabetes in pregnancy.
- Review medical and obstetric history.
- Advise on glycaemic control to optimise HbA_{1C} in women with pre-pregnancy diabetes.
- Review medications in women with pre-pregnancy diabetes.
- Screen for and manage complications related to existing diabetes or gestational diabetes.
- The complications of diabetes in pregnancy, should be explained to the patients (see Table 5.2(1) below).
- Physical activity recommendations to prevent GDM in non-diabetic patients:
  - Pre-conception physical activity reduces the risk of GDM[25].
  - In-pregnancy provision of 30-50 minutes of moderate intensity physical activity in-facility at least three times a week for those at higher risk for GDM (Table 6.2 for the definition of high risk) can help to avert GDM development[26] **[L1]**. This should start before the 20th week of gestation[26].
  - Physical activity is also an important part of a healthy pregnancy, and once the woman becomes pregnant, regular participation should be encouraged[27].
  - Lifestyle counselling in pregnancy does not reduce the incidence of GDM[28-31] **[L1]**.

NB: Three to six months' attendance for pre-pregnancy care is typically required to optimise glycaemic control and address all other issues **[R-GDG]**.

5.2 Complications of Diabetes in Pregnancy

Pre-existing diabetes and gestational diabetes are associated with short-term and long-term complications for both the mother and the baby.

The table below outlines the possible peri-pregnancy complications to both mother and child as a result of maternal diabetes in pregnancy[3,13].

Maternal Complications	Fetal or Neonatal Complications

• Metabolic complications:
  • Elevated average plasma glucose.
  • Unmet glycaemic targets.
  • Adverse events related to therapy (hypoglycaemia and others).
  • Weight gain in pregnancy.
 	
• Pregnancy related complications:
  • Assisted labour/ Delivery (including caesarean).
  • Pre-term delivery.
  • Peri-partum infection.
  • Pregnancy induced hypertension or eclampsia/pre-eclampsia.
	• Fetal or neonatal complications:
	  • Birth weight abnormalities (LGA or SGA).
	  • Hypoglycaemia within one hour of birth.
	  • Neonatal death.
	  • Stillbirth.
	  • Neonatal Intensive Care Unit (NICU) admission.
	  • Shoulder Dystocia.
	  • Respiratory Distress Syndrome (RDS).
	
	Note: The care team should look out for diabetes complications in women with pre-existing diabetes, such as retinopathy (new / worsening), nephropathy (new/worsening) or diabetic keto-acidosis (DKA). In addition, congenital malformations increase if pre-existing diabetes is uncontrolled during early pregnancy (see Section 5.4).

Table 5.2(1): Maternal and Fetal/Neonatal Complications of Diabetes in Pregnancy (Adapted from[3,13,32]).

Serious clinically significant hypoglycaemia is defined as a blood glucose (BG) of <3.0 mmol/L (540 mg/dL), while the BG alert value is defined as ≤3.9 mmol/L (70 mg/dL)[33].

The table below outlines the thresholds for the classification of hypoglycaemia[33].

Level	Glycaemic Criteria	Description
Level 1: Glucose Alert Value	≤3.9 mmol/L (70 mg/dL)	Sufficiently low glucose for treatment with fast-acting carbohydrate and dose adjustment of glucose-lowering therapy.
Level 2: Clinically Significant Hypoglycaemia	<3.0 mmol/L (54 mg/dL)	Sufficiently low to indicate serious, clinically important hypoglycaemia.
Level 3: Severe Hypoglycaemia	No specific glucose threshold	Hypoglycaemia associated with severe cognitive impairment requiring external assistance for recovery.

Table 5.2(2): Classification of hypoglycaemia[33].

The long-term complications of diabetes in pregnancy include increased incidence of childhood adiposity and development of IGT in children aged 10 – 14 years compared with children born to mothers without GDM[34,35]. GDM is also a strong marker for future maternal development of type 2 diabetes mellitus[36]. Other complications may also exist.

5.3 Contraception Advice

Discuss family planning[2]**[L1]**:

- Avoidance of unplanned pregnancy should be an essential component of diabetes education from adolescence for all women with diabetes[3].
- The need for preparation for pregnancy and pre-conceptual counselling should be emphasised during each and every diabetes annual review appointment.

- Prescribe contraception until HbA_{1C} has been optimised [2,3] **[L1]**:
    - Choice of contraception should be based on the patient's preferences and risk factors.
    - Oral contraceptives may be used in the absence of contraindications e.g. cardiovascular disease, hypertension, proliferative retinopathy or nephropathy.
        - In such patients, consider the use of the progesterone-only pill or *Mirena* coil.
- Before discontinuing contraception, offer the patient pre-conception care and advice as above[3].

5.4 Pre-existing Diabetes and Medication Review

Emphasise the importance of glycaemic control:

- Aim to keep HbA_{1C} <6.5%, prior to conception, if it can be achieved without problematic hypoglycaemia[2,3] **[L2, RGA]**.
    - Good glycaemic control is necessary to reduce the risk of congenital abnormalities, miscarriage, stillbirth and neonatal death[2,3].
    - However, explain that risks can be reduced but not entirely eliminated[3].
- If HbA_{1C} is ≥10%, the absolute risk of congenital malformation increases significantly, and women should therefore be advised not to become pregnant.
- The risk of malformations may also be increased at lower HbA_{1C} levels **[R-GDG]**.
- The patient should be informed of the associated risks should pregnancy occur (see Table 5.2(1) above)[3].
- Patients with T2DM should be managed with[2,3]:
    - Diet alone; or
    - Diet plus Metformin; or
    - Metformin plus Insulin.

NB: Metformin has been shown to be safe and efficacious during pregnancy [4-6] ,whilst the safety profile of other oral hypoglycaemic agents is not known and should be stopped prior to conception[7-10].

Insulin Use in Pregnancy[3,7-11]**:**

- Rapid-acting insulin analogues– aspart, ultra-fast aspart (Fiasp) and lispro, are approved in pregnancy and should be continued. They are more advantageous than soluble human insulin during pregnancy in reducing the risk of hypoglycaemia.
- Intermediate-acting insulin (Neutral Protamine Hagedorn (NPH), basal insulin detemir and insulin degludec are both approved in pregnancy.
- Initiation of insulin glulisine or glargine during pregnancy is not recommended. Patients whose diabetes are well controlled on these analogues, do not need to discontinue it during pregnancy[8] **[L2, RGA]**.
- Continuous subcutaneous infusion of insulin (insulin pump) may be considered for those patients otherwise unable to meet glycaemic targets. Patients should be referred to an endocrinologist who is experienced in pump management **[R-GDG]**.

NB: Insulin does not cross the placenta.

Other Medications[2,3,13,21]**:**

- Angiotensin Converting Enzyme (ACE) inhibitors, and Angiotensin Receptor Blockers (ARBs) and Renin-Angiotensin-Aldosterone System (RAAS) blocking drugs[37] should not be used during pregnancy. They should be discontinued before conception or as soon as pregnancy is confirmed[2] **[L1, RGC]**. Alternatives include: Labetalol, Nifedipine and Methyldopa[3] **[L2, RGA]**.
- Statins can be continued during pregnancy in women with familial hyperlipidaemia, those with established cardiovascular disease and those at high risk of cardiovascular disease.
- Hydrophilic statins, such as rosuvastatin and pravastatin, are preferred to lipophilic statins, such atorvastatin and simvastatin's, because of a lower risk of SGA.

5.4.1 Monitoring Blood Glucose in the Pre-Conception Period

Advise the patient on the following[3,13]:

- Newer reliable glucometers or recently calibrated glucometers should be used, and the patient's monitoring technique should be reviewed **[R-GDG]**.
- To record their Blood Glucose (BG) measurements at a minimum of 3-4 times per day up to a maximum of 6-7 times per day.
- The ideal frequency of BG measurement will depend on the insulin regimen used and the frequency of hypoglycaemia. At a minimum, this should include fasting measures and two hours after meals.
- If nocturnal hypoglycaemia is suspected, patient should be advised to measure her BG levels during the night.
- CGM could be used in those with insulin treated diabetes[16].
  - The target glucose should be set between 70-180 mg/dl (3.9-10 mmol/l).
  - Time above Range (TAR) should be ≤ 25%.
  - Time in Range (TIR) should be ≥ 70%.
  - Time below Range (TBR) should be < 4%.
- See *Section 7.5* for glycaemic targets in pregnancy.

5.5 Nutritional Advice and Weight Management

Nutritional advice:

- It is good clinical practice to provide dietary advice before, during and after pregnancy. The diet should be based on low glycaemic index foods which are not excessive in fat[3].
- Women with high Body Mass Index (BMI) and overweight should be encouraged to achieve a normal (BMI) when reasonable[3,38] or significant weight reduction, such as 5-10% of body weight **[R-GDG]**, prior to pregnancy.
- Women with diabetes who are planning to become pregnant should be advised to take high-dose folic acid (5 mg/day) starting at least 3 months prior to conception and continuing until 12 weeks of gestation, to reduce the risk of neural tube defect[3] **[L2]**.

5.6 Physical Activity and Lifestyle Counselling

Physical activity in patients with pre-existing diabetes[2]:

- At least 150 mins per week of moderate-intensity aerobic exercise (50-70% of maximum heart rate) is recommended:
  - Spread-out over at least three days per week.
  - Ensure there are no more than two consecutive days without exercise.
- In patients with T2DM, if there are no contraindications, resistance training is recommended twice a week.
- Patients with type 1 diabetes (T1DM), should be advised about[12]:
  - Safe pre-exercise BG levels (typically ≥100 mg/dL depending on the individual and type of physical activity); and
  - Appropriate adjustment of insulin and meals/snacks to reduce hypoglycaemia.
  - Having a simple carbohydrate food readily available before, during, and after exercise for prevention or treatment of hypoglycaemia.

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5.7 Screening for Complications

Nephropathy[2,8,39,40]:

- There is a strong association between pre-existing nephropathy and a poor pregnancy outcome.
- Worsening nephropathy and superimposed pre-eclampsia are amongst the most common causes of pre-term delivery in women with diabetes.
- The following tests should be undertaken in all women with diabetes:
    - Albumin-creatinine ratio (ACR).
    - Serum creatinine and eGFR.
- Patients with an abnormal ACR, confirmed on repeated testing, should be referred to a diabetologist for specialist review **[R-GDG]**.
- If the eGFR is <40 ml/min/m^3 prior to conception, the woman may experience irreversible further decline in renal function as a consequence of the pregnancy[20]. Such patients should be referred to a nephrologist for review **[R-GDG]**.

Retinopathy[2,3,41]:

- Retinopathy can deteriorate significantly during pregnancy.
- Fundus examination is advised prior to conception, if not performed in the last 6 months.
- Patients with active retinopathy should be under the care of an ophthalmologist **[R-GDG]**.

6 Screening for Diabetes in Pregnancy

6.1 Uncertainties in Screening for Gestational Diabetes Mellitus:

Screening for diabetes in pregnancy is important to diagnose cases with GDM or not yet diagnosed type 2 diabetes. The timing of screening, who to screen, and the method of screening have been a subject of debate.

- Conventionally, screening for GDM is usually undertaken between 24-28 weeks of gestation, though it may be done earlier in pregnancy.
    - The evidence for early screening (12-20 weeks)[42] is insufficient to assess the balance of benefits and harms of screening for gestational diabetes in asymptomatic pregnant women < 24 weeks of gestation[43-50]. Evidence of the effectiveness of treatment of GDM diagnosed early from the TOBOGM trial was not conclusive[51].
    - Given this uncertainty, these guidelines recommend screening early, using the algorithms in *Section 6.2.5*, given the fact that GDM, pre-diabetes, and type 2 diabetes have a high prevalence in Qatar **[R-GDG]**.
- There is currently no data to support universal screening for GDM after 32 weeks gestation, as there are no proven benefits of interventions for either the mother or the offspring.
- The decision to screen for GDM after 32 weeks of gestation, should be individualised to the patient by the treating physician based on the presence of risk factors and/or features to support diabetic fetopathy; and whether the diagnosis of GDM will make any difference to the outcomes **[R-GDG]**.
- Options for screening may include venous fasting BG alone with or without BG testing 2-hours after food. If these are normal, then self-monitoring of blood glucose (SMBG) over a four-day period, could be used in high-risk subjects **[R-GDG]**.

6.2 Screening for Gestational Diabetes

6.2.1 Early Screening (First Trimester)

There is currently no clear guidance that indicates where on the glycemic spectrum (i.e. where on the FPG continuum below 7 mmol/L or 2-hour PG continuum below 11.1 mmol/L) diagnosis of first trimester GDM should be made. Alternatives proposed include the IADPSG/WHO2013 thresholds[49,50] or 'pre-diabetes' HbA_{1C} thresholds of 5.7 to 6.4% at <14 weeks[45,46] or HbA_{1C} thresholds of 6.0 at less than 14 weeks.

- Early screening should follow the algorithms in *Section 6.2.5* given the uncertainties in the literature noted above **[R-GDG]**.
- In the first trimester, FPG and/or HbA_{1C} in the upper half of the ranges for GDM indicated in *Section 6.2.5*, may sometimes be associated with prior undiagnosed diabetes discovered in early pregnancy[52].
- If HbA_{1C} >6.5%, diagnosis of pre-existing diabetes is met, and Oral Glucose Tolerance Test (OGTT) is not needed.

6.2.2 Usual Screening (24 to 28 Weeks)

- At 24-28 weeks, the IADPSG/WHO2013 thresholds[49,50] should be used on a 2-hour oral glucose tolerance test to diagnose GDM (diagnosis if FBG ≥5.1 or 1h BG ≥10 or 2h BG ≥8.5 mmol/L).
    - Screening in patients with a history of bariatric surgery (less than 18 months from surgery date), should be performed using fasting glucose and HbA_{1C} as the OGTT is not reliable in this group **[R-GDG]**.

6.2.3 Risk Stratification

A precision medicine approach to GDM which recognizes severity may help individualize treatment strategies, leading to improved clinical care and outcomes[53].

- A broad risk grouping are class A1 (glycemic control achieved without medication) and class A2 (glycemic control achieved with medication)[2,54]. The weighted average glucose (dwAG) threshold from the NPRP criteria for GDM between 24 to 28 weeks of pregnancy may be useful to assist physicians in making more informed risk stratification **[R-GDG]**[55,56].

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dwAG*	Status
≤ 6.8 mmol/L	Normal gestational glycaemia **(NGG)**
> 6.8 ≤ 7.5 mmol/L	Impaired gestational glycaemia **(IGG)**
>7.5 ≤ 8.6 mmol/L	Gestational diabetes mellitus **(GDM)**
>8.6 mmol/L	High-risk gestational diabetes mellitus **(hGDM)**

* dwAG is computed as (fasting plasma glucose×0.28) +(1hr plasma glucose×0.36) +(2hr plasma glucose×0.36).

6.2.4 Screening Tests Summary

Screening tests to detect undiagnosed T2DM or GDM should be undertaken as described in the below table **[R-GDG]**:

Target population	Time of screening	Test*
Average-risk women: • All pregnant women without the risk factors listed for high-risk women below.	At booking visit	**Fasting BG**
	At 24-28 weeks	If the booking visit test is normal: Perform 75g OGTT

High-risk women:
Women with any of the following should be considered to be at high risk of having undiagnosed diabetes or for developing GDM in the current pregnancy:
• Women with a history of GDM.
• Women with a history of impaired fasting glucose and/or impaired glucose tolerance.
• History of unexplained stillbirth, IUFD or unexplained neonatal death or birth of baby with malformations associated with diabetes.
• History of a macrosomic baby weighing ≥4kg.
• Women with the following pre-pregnancy results:
  ◦ **HbA_{1c}**: 5.7% - 6.4%.
  ◦ **Fasting BG:** 5.6 - 6.9 mmol/L (100-125mg/dl).
  ◦ **OGTT at 2hrs**: 7.8 - 11.0 mmol/L (140-199 mg/dl).
• Pre-pregnancy BMI of >25 kg/m2 (have a lower threshold in South Asian women).
• Women with a history of PCOS.
• Other rare high-risk patient e.g. long-term steroid users.
	At booking visit	Fasting BG and HbA_{1C}**

	At 16-18 weeks	If the booking visit tests are normal: Perform 75g OGTT
	24-28 weeks (preferably at 24 weeks)	If the prior tests are normal: Perform 75g OGTT

Table 6.2.4: Screening Algorithm for Diabetes in all Pregnant Women (Adapted from[13]).

* In women who have excessive vomiting in early pregnancy, fasting BG may be unreliable. Ensure to confirm no vomiting prior to test to avoid false positive results **[R-GDG]**.

Note that pregnant women may have their first booking visit at any trimester, and screening using HbA_{1C} is not suitable after 14-weeks **[R-GDG].

6.2.5 Screening Algorithms

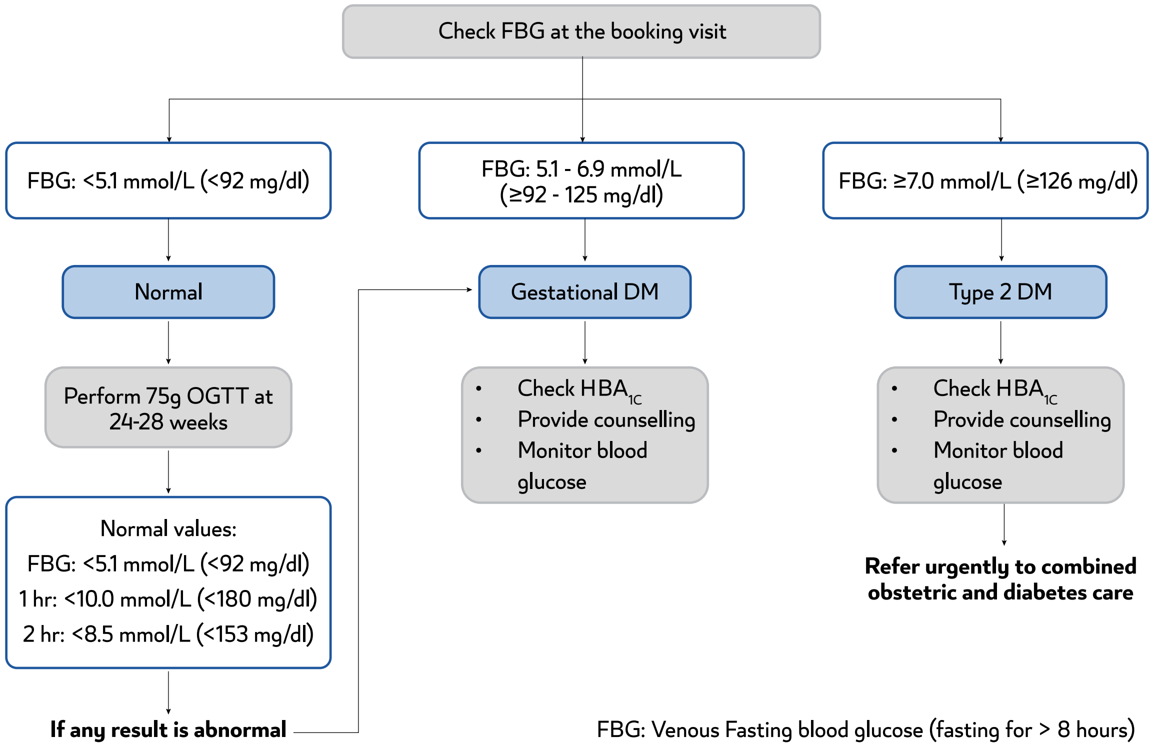

Fig 6.2.5 (1): Screening and Diagnosing GDM [15,57,58] and T2DM in Average-Risk Women [Adapted from[13]].

NB: Patients who do not tolerate OGTT should be screened with serum fasting BG and capillary SMBG for three consecutive days **[R-GDG].**

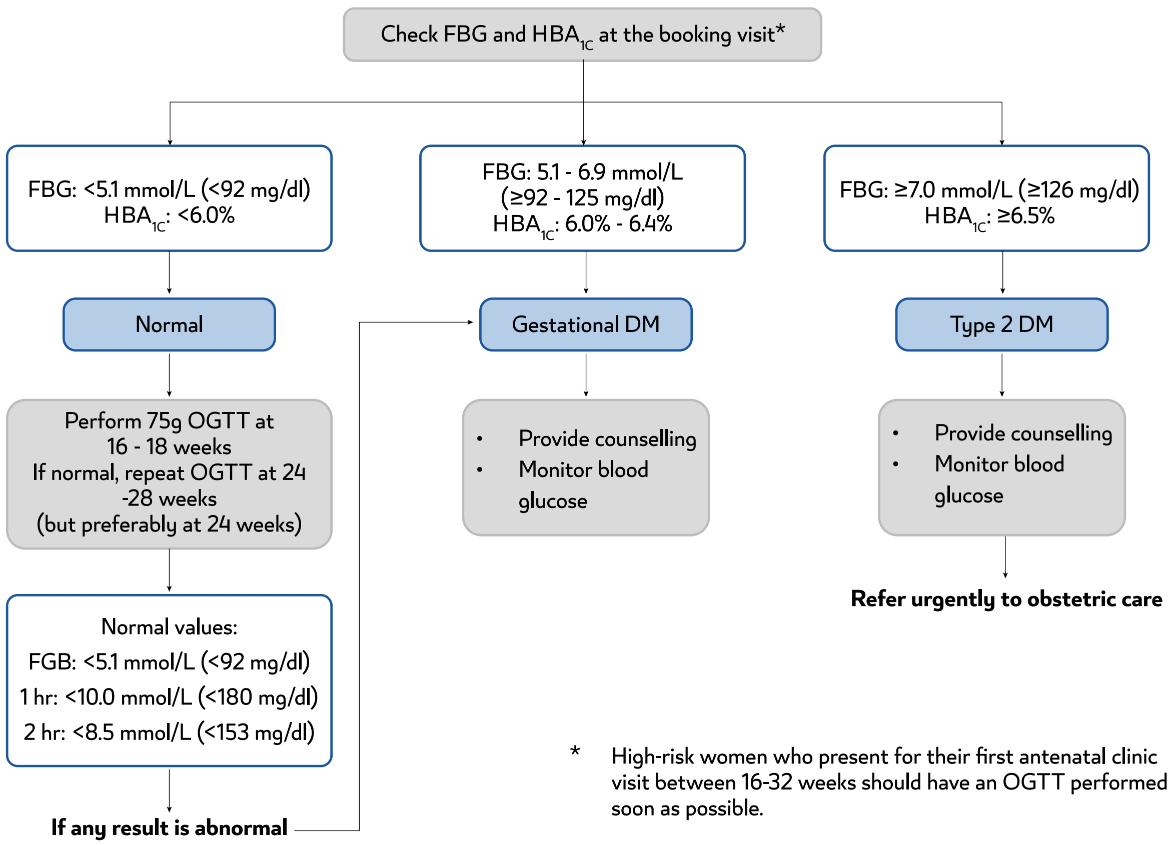

Fig 6.2.5 (2): Screening and Diagnosis of GDM and T2DM in High-Risk Women [Adapted from [13]].

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6.2.6 Screening Women with Bariatric Surgery:

- OGTT should be avoided in women who underwent bariatric surgery ≤18 months.
- In the first trimester, screen with FBG and HbA_{1C}.
- Between 16-18 weeks of gestation, screen using fasting glucose. If normal, then screen using a 4 points SMBG using a calibrated glucometer for 7 days. Women should monitor the FBG and the 1-hour glucose post meals. The glucose targets are FBG <5.3 mmol/l (95 mg/dl) and the 1-hour post meal glucose <7.8 mmol/l (<140mg/dl). GDM is diagnosed if ≥20% of the readings are above targets.
- Repeat the same process between 24-28 weeks of gestation if the above tests are normal.

6.2.7 Pitfalls in Early Screening Results

- Elevated FBG should be repeated in the following conditions.
  - Women with hyperemesis gravidarum.
  - Women with thyrotoxicosis.
  - Women with Polycystic Ovary Syndrome (PCOS) who are already onMetformin should continue Metformin till 24 weeks. Metformin should be held thereafter for one week, and the women can undergo OGTT.
- HbA_{1C} has poor sensitivity for detecting GDM. Hence it should be used as an adjunct screening tool with venous glucose.
- Capillary glucose readings have high variability and are less accurate than venous glucose values, and hence should not be routinely used for screening for GDM.

6.3 Referral Criteria to Specialist Care

Referral to the Emergency Department [R-GDG]:

- If signs of DKA, hyperglycaemic hyperosmolar state (HHS) or persistent hyperglycaemia despite treatment, are present.
- If ultrasound in the third trimester shows intrauterine growth restriction (IUGR) or fetal compromise.

Urgent Referral to Specialist Antenatal Care [R-GDG]:

- A pregnant woman who is a known diabetic or is diagnosed as T2DM during her first screening.
- GDM patients who do not meet glycaemic targets within two weeks with dietary management (i.e. ≥20% of readings outside the normal range).
- Patients with GDM who have high BMI ≥40kg/m^2.
- GDM patients who are ≥28 weeks gestation and who have two hours postprandial glucose levels of ≥10 mmol/L (180mg/dL) on three consecutive days require urgent referral to a specialist physician within a maximum of three days. If this is not possible, the patient should be sent to the emergency room to seek advice from an endocrinology consultant.

Routine Referral to Specialist Care:

- Refer all patients with pre-gestational diabetes to an ophthalmologist for retinal assessment **[R-GDG]**.
- All patients whose pregnancy GTT has the risk stratum "Gestational Diabetes" or "High-Risk Gestational Diabetes" according to National Priorities Research Program (NPRP) criteria [56] will be routinely referred to specialist care.

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7 Management of Gestational Diabetes Mellitus

7.1 Antenatal Care

Antenatal care for women with GDM should be provided by a multidisciplinary team consisting of **[R-GDG]**:

- Family physician/obstetrician and endocrinologist if needed.
- Health counsellor/educator.
- Dietician.

Note[3]:

- The frequency of antenatal visits is decided on an individual basis, typically every 1-3 weeks after diagnosis.
- At each visit the following should be checked, in addition to routine antenatal care:
  - Blood pressure.
  - Urine dipstick primarily for proteinuria.
  - Body weight.
  - Review of SMBG.

7.2 Education

All patients with GDM should receive counselling and education (supported by an information leaflet) that explains the following **[R-GDG]**:

- The risks to both mother and baby, associated with GDM in pregnancy (see Table 5.2(1)).
- The symptoms of hypoglycaemia.
- Diet, exercise and weight gain during pregnancy.
- The importance of good maternal glycaemic control to avoid complications.
- Education about oral hypoglycaemic agents (including action, side effects, etc.) and insulin injection when needed, e.g., when diet management alone is insufficient.
- The possibility of transient morbidity in the baby during the neonatal period, which may require admission to the neonatal unit.
- The risk of the baby developing obesity and/or diabetes in later life.
- The benefits of exercise, in the absence of contraindications of 30 minutes per day (as a single session or as 10 minutes after each meal)[59]

Change in lifestyle is essential in the management of women with GDM, and in some cases it may be the only treatment required [2,3] **[L1]**.

7.3 Weight Management

Women should adhere to the following acceptable weight gain limits during pregnancy in order to minimise the risk of complications, but active weight loss should not be attempted **[R-GDG]**.

Weight category	Pre-Pregnancy BMI	Acceptable weight gain by gestation			
		<20 weeks		≥20 weeks	
Underweight	<18.5	4.0 - 6.0kg	(0.2-0.30 kg/wk)	8.0 - 12.0kg	(0.4-0.6 kg/wk)
Normal	18.5 - 24.9	4.0 - 5.5kg	(0.2-0.25 kg/wk)	7.5 - 10.5kg	(0.35-0.5 kg/wk)
Overweight	25.0 - 29.9	2.0 - 3.0kg	(0.1-0.15 kg/wk)	5.0 - 8.0kg	(0.25-0.4 kg/wk)
Obesity Class I	30.0 - 34.9	0kg		5.0 - 8.0kg	(0.25-0.4 kg/wk)
Obesity Class II & III	≥35	0kg		0 - 5.0kg	(0-0.25 kg/wk)

Table 7.3: Acceptable Weight Gain for Women with GDM by Pre-Pregnancy BMI and Gestational Age [60].

Caloric intake[8,27]:

- Is the best predictor of weight gain during pregnancy.
- Calculate the appropriate caloric intake based on the patient's pre-pregnancy BMI, her current pregnant BMI and the acceptable weight gain during pregnancy.
- Caloric intake should increase in the second and third trimesters[27] **[L3, RGA]**:
- Additional calorie consumption in pregnant with high BMI may not be necessary[33]:
    - Woman with high BMI may reduce her calorie intake during pregnancy by 30% compared to their pre-pregnancy intake[8] **[L1, RGA]**, with a minimum intake of 1600[8,13,61].

Dietary advice:

- Offer advice about diet and exercise at the time of GDM diagnosis[3] **[L1, RGA]**.
- Advise the patient to eat a healthy diet during pregnancy[3] **[L1, RGA]**.
- Emphasise the benefit of foods with a low glycaemic index[3] **[L1, RGA]**.
- All patients with GDM should be referred to a dietitian for dietary planning[2,3] **[L2]**.
- All women with GDM should adhere to the healthy eating recommendations. [27] **[L2]**.
- Advise the patient to eat small, frequent meals rich in protein to reduce the risk of postprandial hyperglycaemia and pre-prandial starvation ketosis[27] **[L2]**.
- Advice should be tailored to each patient's individual needs, culture, and preferences[27] **[L2]**.

Carbohydrate intake:

- It is recommended that women with GDM limit carbohydrate intake to 35% - 45% of total calorie intake[3,13,61] with a minimum of 175 g carbohydrate per day distributed in three small to moderate sized meals and two to four snacks **[R-GDG]**.
- Limit breakfast carbohydrate amounts to as little as 15-30 g, due to hormonal surges and insulin resistance in the morning hours. Fruit and milk should be avoided at breakfast **[R-GDG]**.
- Ensure to add evening snacks to avoid ketosis overnight. A bedtime snack (if recommended) should include one to two servings of high-fibre carbohydrates with one to two servings of protein. Evaluate the need for this snack based on fasting blood sugar levels **[R-GDG]**.
- Patients should consider meeting carbohydrate requirements using the following foods[27]:
    - Vegetables.
    - Legumes.
    - Whole grains.
    - Complex carbohydrates.
- Discourage the consumption of simple carbohydrates[27].
- Substituting low-glycaemic load foods for higher-glycaemic load foods as the carbohydrate effect on blood glucose differs among types of food. Low glycaemic index foods often have higher fibre content. Up to 28g of fibre intake per day is recommended for pregnant women **[R-GDG]**.

Protein[27]:

- Woman who is pregnant should consume around 1.1 g/kg of protein per day in the second and third trimesters[27] **[L3, RGA]**.
- At minimum, protein should account 15-20% of total calories [62]. 71 g of protein per day is recommended **[R-GDG]**.
- Non-pregnant woman requires (0.8 g/kg/day).
- Patients should consider meeting protein requirements from animal, fish, or plant sources.

Fat[27]:

- Fact should account 30-40% of total calories[63] and less than 10% of calories should be from saturated fats[64].
- Trans-fatty acids should be avoided.
- The diet should contain 13 g/day of omega-6, and 1.4 g/day of omega-3.

Exercise[65]:

- Apart from contact sports, women can continue their usual exercise program during pregnancy.
- For those who were not involved in exercise, a minimum of 150 minutes of aerobic exercise per week, such as walking, stationary cycling, and dancing is recommended.
- Women can split the daily 30 minutes exercise sessions into three 10 minutes sessions post meals.
- One to two sessions of resistance training per week (using bands or weight) is recommended.
- Women with the following symptoms should not exercise:
    - Vaginal bleeding.
    - Abdominal pain.
    - Regular painful contractions.
    - Amniotic fluid leakage.
    - Dyspnoea before exertion
    - Dizziness.
    - Headache.
    - Chest pain.
    - Muscle weakness affecting balance.
    - Calf pain or swelling.

7.4 Home Glucose Monitoring

Home Glucose Monitoring:

- Every patient should have a reliable and recently calibrated glucometer at home[3].
- Patients on diet management alone, Metformin therapy or single dose of basal insulin, should be advised to monitor their BG levels, four times per day, including[3]:
    - Fasting; and
    - One hour or two hours post-meals.
    - If glucose levels are in the target range after two weeks of four times daily glucose testing, every other day testing may be recommended[66].
- Patients on multi-dose injections of insulin should be advised to monitor their BG levels, up to six to seven times per day to include[3,11]:
    - Fasting.
    - Pre-meals.
    - One hour or two hours post-meals.
    - Before bedtime.

7.5 Glycaemic Targets

- Glycaemic Targets should be individualised taking into account the risk of hypoglycaemia. Patients on insulin should keep their capillary BG >4 mmol/L (72 mg/dl)[3]**[L3]**.
- If achievable without causing problematic hypoglycaemia, control should aim for[3,11,13,15] **[L1, RGA]**:
    - Fasting BG should be ≤5.3 mmol/L (<95 mg/dL) [2,3,11,13,61,63,64,67,68].
    - 1 hour after meals ≤7.8 mmol/L (140mg/dL) [2,3,11,13,61,69].
    - 2 hours after meals ≤6.7 mmol/L (<120 mg/dL) [11,13,58,61,63,64,67,68,70].
    - Post-prandial rather than pre-prandial glucose levels are associated with better neonatal and maternal outcomes[71,72].
    - Evidence for using tighter post-prandial targets is insufficient [73].
    - Glucose values 20-30% above target are suboptimal **[R-GDG]**.

- In women on CGM **[R-GDG]**
  - The target glucose should be set between 63-140 mg/dl (3.5-7.8 mmol/l).
  - Time above Range (TAR) should be ≤ 15%.
  - Time in Range (TIR) should be ≥ 80%.
  - Time below Range (TBR) should be < 4%.
- Regular monitoring of HbA_{1C} is not recommended after measurement at initial diagnosis[3] **[L2]**.

7.6 Fetal Surveillance

- Ultrasound surveillance during the 3rd trimester, starting from week 28-32 and every three to four weeks until delivery is recommended in order to estimate the effect of glycaemic control on fetal growth and amniotic fluid volume[69].
- The frequency and methods of fetal monitoring are determined by maternal glycaemic control and the existence of other pregnancy complications **[R-GDG]**.
- Biophysical profile testing and multi-vessel Doppler studies may be considered in cases of Intrauterine Growth Restriction (IUGR) or when there are co-morbid conditions, such as pre-eclampsia **[R-GDG]**.

For women diagnosed with GDM in the first trimester (undiagnosed pre-existing diabetes should be considered), fetal surveillance includes[3]:

- A viability scan and confirmation of gestational age, preferably at seven to nine weeks, but no later than 12 weeks, where possible.
- Nuchal translucency scan at 11-13 weeks.
- Fetal anomaly scan at 18-20 weeks.
- Fetal growth and amniotic fluid volume assessment at 28-32 weeks and repeated at 36 weeks.
- Further scans should depend upon glycaemic control and clinical judgment.

In the event of identified macrosomia, a clear management plan should be drawn by a consultant obstetrician to address timing of follow-up scans, fetal surveillance, timing and mode of delivery[3].

Women with GDM should be advised to monitor fetal movements from 24 weeks and report any concerns immediately to the healthcare team **[R-GDG]**.

7.7 Pharmacotherapy for GDM

Management of blood glucose:

- Diet and physical activity should be the first step in management[2,3].
- If patients fail to achieve the glycaemic targets outlined in *Section 7.5* within one to two weeks, then medication should be commenced[58,63,64].
- Most GDM guidelines[42,74] recommend insulin as the preferred treatment or first-line therapy for persistent hyperglycaemia when lifestyle measures are insufficient to achieve good glycaemic control.
- Recent data suggests that Metformin and glibenclamide are at least equivalent to insulin for choice of initial therapy. Metformin found to be the optimal choice of the two medications when each compared against insulin in terms of less hypoglycaemia and hypertensive disorders in pregnancy[14] **[L1]**.

7.7.1 Metformin

Metformin[5,14]

- Should be offered:
  - To patients with GDM as first line if BG targets are not being achieved using diet and exercise within 1-2 weeks[2,3,14] **[L1, RGA]**.
- May be preferred to insulin for maternal health if it controls BG sufficiently[2,14] especially in women who are [3]:
  - Declining insulin therapy.
  - Not affording insulin therapy.

    - Unable to safely use insulin.
    - With suspected severe insulin resistance.
- Patients should be advised that[2] **[L2]**:
    - Oral agents cross the placenta.
    - No adverse effects on the fetus have been demonstrated, however long-term studies are in progress.
- Commence treatment at 500 mg BD and titrating to a dose of 1g BD over three to seven days, as tolerated **[R-GDG]**.
- Contraindications should be carefully reviewed before prescribing **[R-GDG]**.
- Metformin should not be initiated in patients with the following conditions and must be stopped if already initiated [2,75]**[L1, RGC]**:
    - Renal dysfunction eGFR < 60 ml/min/1.72 m^2) (check with the previous statement above).
    - High liver enzymes.
    - Pre-eclampsia.
    - Patients at risk of or with actual IUGFR.
- For more details, review the NCG on The *Management of Obesity in Adults* by MoPH available at MoPH website via Ministry of Public Health - Clinical Guidelines[76].

7.7.2 Insulin

Offer insulin treatment:

- Insulin should be started if Metformin is contraindicated or is not acceptable or tolerated by patients[3]**[L1, RGA].**
- In addition to Metformin and diet and lifestyle changes if BG targets are not achieved[3] within one week on maximal doses of metformin **[R-GDG]**.
- Refer to *Section 5.4* for further detail on the types of insulin that may be used in pregnancy.

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8 Management of Type 1 and Type 2 Diabetes in Pregnancy

8.1 Nutrition and Diet

Pregnant women with diabetes should follow the same nutritional vitamin and mineral advice as for women without diabetes in pregnancy[3].

8.2 Glucose and HbA$_{1C}$ Monitoring

Home Glucose Monitoring:

- Every patient should have a reliable and recently calibrated glucometer at home[3].
- Patients on diet management alone, Metformin therapy or single dose of basal insulin, should be advised to monitor their BG four times per day, including[3]:
  - Fasting
  - One hour or two hours post-meals.
  - Target glucose one-hour post-meal < 7.8 mmol/L and two-hours post-meal < 6.7 mmol/L as there are no Randomized Controlled Trials (RCTs) to inform on the best glycaemic target **[R-GDG]**
- Patient on multi-dose injections of insulin should be advised to monitor their BG levels, up to six to seven times per day to include[3,11]:
  - Fasting.
  - Pre-meals.
  - One hour or two hours after meals.
  - Before bedtime.

Continuous Glucose Monitoring System (CGMS):

- CGMS should be offered to:
  - All women with DM-1[5] **[L1]**.
  - All women with DM-2 on insulin.
  - All women with DM-2 post bariatric surgery.
  - Women who have unstable BG levels.
  - It should also be offered to the woman if self-monitoring of BG has failed to assess glycaemic control; or to gain more information about variability in BG.

- CGMS may help to increase time in target and reduce hypoglycaemia, but information on the impact on maternal and fetal outcomes are limited and inconsistent[77-79].

Hypoglycaemia:

The following are essential[3]:

- All patients should be advised about the risks of hypoglycaemia, and impaired awareness.
- Patients and at least one family member, ideally the husband, should be educated regarding management of hypoglycaemia.
- Women with insulin-treated diabetes should always have a fast-acting form of glucose available to them **[L2]**.
- Provide glucagon to pregnant women with T1DM for use if needed **[L2]**.
  - A family member should be educated on when and how to use glucagon.

HbA$_{1C}$ Monitoring:

- HbA$_{1C}$ should be measured in the first clinic review[3,11].
- Due to increases in red blood cell turnover associated with pregnancy, HbA$_{1C}$ levels fall during pregnancy. HbA$_{1C}$ should be used as a secondary measure and should not replace SMBG[15].

- HbA_{1C} should be measured at least once in each trimester or more frequently e.g. monthly[2,3].
- The frequency of monitoring should be judged by the treating physician **[R-GDG]**.

8.3 Glycaemic Targets

Glycaemic Targets:

- Should be individualised taking into account the risk of hypoglycaemia[3].
- Patients on insulin should keep their capillary BG >4 mmol/L (72 mg/dl)[3].
- If achievable without causing problematic hypoglycaemia[3,11,15] **[L1, RGA]**:
    - Fasting BG ≤5.3 mmol/L (≤95 mg/dL).
    - One hour after meals ≤7.8 mmol/L (140mg/dL).
    - Two hours after meals ≤6.7 mmol/L (≤120 mg/dL).
- Due to increased red blood cell turnover during pregnancy the HbA_{1C} target is lower:
    - Use ≤6% if this can be achieved without significant hypoglycaemia[2]
    - Relax target to between ≤6.5% to ≤7% if necessary to prevent hypoglycaemia[2]
- Continuous glucose monitoring should be offered to all women with type 1 DM on insulin[16]:
    - The target glucose should be set between 63-140 mg/dl (3.5-7.8 mmol/l).
    - Time above Range (TAR) should be ≤ 25%.
    - Time in Range (TIR) should be ≥ 70%.
    - Time below Range (TBR) should be < 4%.
- Continuous glucose monitoring could be offered to women with T2DM treated with insulin **[R-GDG]**:
    - The target glucose should be set between 63-140 mg/dl (3.5-7.8 mmol/l).
    - Time above Range (TAR) should be ≤ 15%.
    - Time in Range (TIR) should be ≥ 80%.
    - Time below Range (TBR) should be < 4%.

8.4 Pharmacological Management

Treatment[3,8,17]:

- A multi-dose injection (MDI) regime is recommended for the majority of patients.
- Patients with T2DM taking Metformin can be continued on their treatment as usual[3,19] **[L2]**.
- Patients should be empowered and taught how to adjust their own insulin doses.
- Frequent telephone contact with the diabetes educators is encouraged for those who need support with dose adjustment.
- A fall in insulin requirement in pregnancy could indicate placental insufficiency. This should not be considered an automatic indication for immediate delivery, but an indication for closer monitoring **[R-GDG]**.

Insulin Pump Therapy in T1DM[3,80]:

- Women with T1DM who are pregnant or are contemplating pregnancy should be considered pre-conceptually for insulin pump therapy, if they fail to achieve their target HbA_{1C}, given that any improvement in control could have significant benefits for fetal and maternal outcomes.
- Women who conceive on MDI should be offered insulin pump therapy during pregnancy if targets for glycaemic control are not achieved (HbA_{1C} <7.0%; BG of 4.4-6.1 mmol/L, before meals, <8.6mmol/L, 2 hours after meals) or if problematic hypoglycaemia occurs[3,16].
- For women using continuous insulin infusion pumps, targets for assessment of glycaemic control during pregnancy are the following[16]:
    - Target range: 63-140 mg/dL and percent of readings: >70%.
    - Below target level: <63 mg/dL and percent of readings: <4%; <54 mg/dL and percent of readings <1%.
    - Above target level: >140 mg/dL and percent of readings <25%.
- The decision whether to continue pump therapy post-partum should be made on an individual basis[80].

8.5 Management of Complications

8.5.1 Vomiting

Patients with diabetes in pregnancy are prone to ketosis in the presence of recurrent vomiting. Such patients should be instructed on how to cope with it and, in cases of severe vomiting, should be hospitalised[3].

Bolus doses of insulin can be given 15-20 minutes after food to avoid hypoglycaemia in cases of recurrent vomiting **[R-GDG]**.

8.5.2 Nephropathy

Pre-existing nephropathy is associated with a poor pregnancy outcome[39]. The incidence of worsening chronic hypertension or pregnancy-induced hypertension/pre-eclampsia is high (40-70%) in women with both incipient and overt nephropathy[3].

All patients should undergo the following[3]:

- Screening for albumin excretion at booking and/or in the first or second trimester.
    - **NB:** Proteinuria increases transiently during pregnancy, returning to pre-pregnancy levels within three months of delivery.

In women with nephropathy, the following is recommended **[R-GDG]**:

- Monitor using a morning urine sample for Albumin-Creatinine Ratio (ACR).
- For high-grade proteinuria (particularly if the patient is suspected of having nephrotic syndrome) a 24-hour urine collection for proteinuria may be requested.

Consider referral to a nephrologist if **[R-GDG]**:

- Serum creatinine is abnormal (≥ 120 µmol/l)
- The urinary ACR is > 30 mg/mmol; or
- Total protein excretion exceeds 2g/day.

8.5.3 Retinopathy

Pregnancy is a risk factor for the development and progression of diabetic retinopathy. This may progress into sight threatening disease and can cause devastating visual impairment. Retinal assessment by an ophthalmologist, should be conducted as follows[3]:

- All women with pre-existing diabetes should have retinal assessment in the first trimester and at 28 weeks.
- Women with diabetes, first diagnosed during pregnancy, should have a retinal assessment performed as soon as possible.
- If retinopathy is noted, then close follow-up throughout pregnancy and for one year postpartum, is required.

NB: Diabetic retinopathy should not delay rapid optimisation of glycaemic control and should not be considered a contraindication to vaginal delivery **[R-GDG]**.

8.5.4 Diabetic Ketoacidosis (DKA) and Hyperglycaemic Hyperosmolar State (HHS)

Women should be educated about DKA and HHS and its prevention through SMBG, appropriate diet, suitable pharmacological therapy and sick-day management. Euglycaemic ketoacidosis is also well recognised in pregnancy the following should be considered [2]:

- Investigation for DKA or HHS, is indicated in women who present with nausea, vomiting, abdominal pain and hyperglycaemia.
- If diagnosed, admit the patient to a high dependency unit in accordance with hospital guidelines and include continuous fetal monitoring after 24 weeks gestation.
- Immediate delivery may not be necessary as fetal heart rate abnormalities may immediately resolve with correction of the metabolic state.

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8.5.5 Thyroid Dysfunction

All women with T1DM should be screened for thyroid dysfunction with thyroid stimulating hormone (TSH) levels and thyroid peroxidase antibodies during the first trimester of pregnancy[12] **[L1, RGA]**.

8.5.6 Pre-eclampsia Prevention

As aspirin can reduce the risk of preterm preeclampsia in women who are at high-risk of preeclampsia, it should be offered 100 mg daily from 13 weeks onward to all women with pre-existing T1DM or T2DM[18] **[L1, RGA]**.

8.6 Fetal Surveillance

Fetal surveillance:

- The frequency and methods of fetal monitoring are determined by maternal glycaemic control and the existence of other pregnancy complications **[R-GDG]**.
- Biophysical profile testing and Doppler velocity studies to assess umbilical blood flow and middle cerebral artery blood flow may be considered in cases of excessive or poor growth or when there are co-morbid conditions, such as pre-eclampsia[3].

For women with T1DM or T2DM diagnosed in pregnancy:

- Fetal surveillance includes[3]:
    - A viability scan and confirmation of gestational age, preferably at seven to nine weeks, but no later than 12 weeks, where possible.
    - Nuchal translucency scan at 11-13 weeks.
    - Fetal anomaly scan at 18-20 weeks.
    - Third trimester ultrasound assessment for:
        - Fetal growth.
        - Assessment of liquor volume and fetal movements.
        - Estimation of fetal growth should be performed on a four-weekly basis from 24 weeks, in order to assess fetal growth velocity.
        - More frequent ultrasound and fetal assessment will be determined by glycaemic control, fetal status or maternal health.

In the event of identified macrosomia:

- Clinicians should be aware that the accuracy of fetal weight estimation is ± 20% in term babies and that accuracy decreases with increasing birth weight **[R-GDG]**.
- A clear management plan should be put in place by a consultant obstetrician. This plan should address timing of follow-up scans, fetal surveillance and timing and mode of delivery[3].
- Sonographic estimation of fetal weight should be combined with the clinical judgment of an obstetrician experienced in the management of pregnancies complicated with diabetes when evaluating the most appropriate mode of delivery for the patient **[R-GDG]**.

Unexpected intrauterine death:

- Remains a significant contributor to perinatal mortality in pregnant women with diabetes[3].
- Conventional tests of fetal wellbeing (umbilical artery doppler ultrasound, cardiotocography and other biophysical tests) have been shown to have poor sensitivity for predicting such events[3].
- It is recommended that patients need to be sensitive in perceiving the fetal movements[3].
- In the event of any changes in the character of the fetal movements (in terms of quantity or quality) the patient should report to the hospital emergency department for CTG and evaluation **[R-GDG]**.

9 Special Considerations

Fasting during Ramadan[81]:

- All pregnant women with either Gestational Diabetes Mellitus (GDM) or diabetes in pregnancy who are managed with insulin therapy or sulfonylureas, are regarded at very high risk of complications and should be strongly advised not to fast.
- Pregnant patients who are managed with either diet alone or Metformin should be considered at high risk and also discouraged from fasting.
- If the patient insists on fasting, ensure the following are provided[81]:
  - Inform patients of risks associated with fasting: Seek specialist advice.
  - Close supervision by a multidisciplinary diabetes team.
  - Information and counselling on frequent SMBG.
  - Information and counselling to adjust medication doses in response to SMBG results.
  - Advice to break the fast if hypoglycaemia or hyperglycaemia occurs.
  - Advice to refrain from continued fasting if frequent hypoglycaemia or hyperglycaemia occurs, or there is a deterioration in other related medical conditions as a result of fasting.

Driving [R-GDG]:

- There are currently no regulations for diabetes and driving in the state of Qatar.
- Patients on insulin treatment during pregnancy are typically on a very intensive regimen, which renders them at higher risk for recurrent hypoglycaemia and hypoglycaemia unawareness.
- Patients who have had one or more episodes of severe hypoglycaemia and/or, are suffering with hypoglycaemia unawareness should be advised not to drive during pregnancy.
- Include the following in advising the patents for safe driving:
  - Check capillary BG prior to starting a journey; do not drive till BG is >5.0 mmol/L.
  - Avoid long journeys or take frequent breaks.
  - Always keep hypo treatment handy in the car.
  - If you feel hypoglycemic; stop the car; take the switch off the ignition; correct the hypoglycemia and do not start driving until 45 minutes after normalising the BG.

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10 Intrapartum Care

10.1 Timing and Mode of Delivery

In all cases of diabetes in pregnancy, delivery should take place in a hospital setting, under the supervision of an obstetrician with a neonatologist available if necessary. The frequency of fetal monitoring should be increased during the intrapartum period.

Women with GDM:

- Advise delivery no later than 40 weeks plus six days, if glycaemic control is satisfactory[3].
  - Offer elective delivery by IOL or caesarean section if delivery has not occurred by then.
- If glycaemic control is unsatisfactory plan for delivery, as for women with T1DM or T2DM in pregnancy, described below.

Women with T1DM or T2DM:

- Advise delivery by induction of labour (IOL) or elective caesarean section (if indicated)[3]:
  - At 38-39 weeks of pregnancy **[R-GDG]**:
    - If there are no metabolic or other maternal or fetal complications and the diabetes is well controlled.
  - Before 38 weeks:
    - If metabolic or other maternal or fetal complications are present (including poor maternal glycaemic control).

Elective Caesarean Section:

- May be considered in women with diabetes and estimated fetal weight ≥4500 grams[42].
- Women should, however, be counselled about the poor predictive ability of ultrasound estimates of fetal weight and the risk and benefits of caesarean delivery[42].
- In the absence of any complications and the presence of good glycaemic control, diabetes is not in itself a contraindication to attempting vaginal birth following previous caesarean section[3].

10.2 Glycaemic Control During Labour

During labour:

- Hourly BG monitoring is recommended, aiming to keep BG between 4-7 mmol/L[3] **[L2]**.
- Patients using insulin:
  - Should be allowed a light diet if desired.
  - Variable rate insulin infusion (VRII), should be considered (as shown in Table 10.2 with a dextrose (substrate) infusion (5% dextrose) + 20mmol/L KCl at 125ml/h) in the following situations:
    - Patients with T1DM.
    - All other patients with diabetes in pregnancy if the capillary BG is ≥7 mmol/L for two consecutive hours.
- For patients who are diet-controlled or using Metformin[3]:
  - VRII should only be commenced if the BG is ≥7 mmol/L for two consecutive hours.

View Pathway

<table>
<tr><th rowspan="2">Blood glucose (mmol/L)</th><th>Protocol A
Insulin rate (ml/hr)</th><th>Protocol B
Insulin rate (ml/hr)</th><th>Protocol C
Insulin rate (ml/hr)</th></tr>
<tr><th colspan="3">Prepare the infusion by adding 50 units of Actrapid to 50 mls of normal saline (0.5 unit/mL). Infuse dextrose 5%/0.45 normal saline at a rate of 100 mls per hour.</th></tr>
<tr><td>≤ 4.0</td><td colspan="3">Stop the insulin infusion for 15 minutes and treat hypoglycaemia, preferably orally, but otherwise with 150-200 mls of 10% dextrose over 15 minutes or by temporarily increasing the glucose infusion rate to 150 mls/min.</td></tr>
<tr><td>4.1 - 7.0</td><td>1</td><td>2</td><td>4</td></tr>
<tr><td>7.1 - 9.0</td><td>2</td><td>4</td><td>6</td></tr>
<tr><td>9.1 - 11.0</td><td>3</td><td>6</td><td>8</td></tr>
<tr><td>11.1 – 13.0</td><td>6</td><td>8</td><td>10</td></tr>
<tr><td>≥ 13.1</td><td>8</td><td>10</td><td>12</td></tr>
</table>

Table 10.2: Variable Rate Insulin Infusion **[R-GDG]**.

Instructions:

- For patients persistently out of range:
  - First ensure proper functioning of the cannula and infusion equipment.
  - Check that the substrate infusion is running at the correct rate.
- If CBG remains >12 mmol/L and is not falling on at least two consecutive measurements, the insulin infusion rate can be increased to the next scale.
  - If the patient is already on C, seek immediate advice from the diabetes team.
  - If CBG is 4–6 mmol/L (on at least two consecutive measurements) insulin infusion rate may need to be decreased or, if this has been done, to increase the substrate to 10% glucose or to seek advice from the diabetes team (including to ensure that a VRII is needed).

10.3 Induction of Labour (IOL)

Instructions for IOL in patients with diabetes in pregnancy (where IOL is indicated)[3]:

- Patients on Metformin should omit their morning dose.
- Patients on basal-bolus insulin should **[R-GDG]**:
  - Receive their basal insulin as normal the night before the IOL.
  - Patients on morning basal insulin doses should receive half their regular dose on the morning of the induction. This is particularly important in patients with T1DM.
  - All other insulin doses should be omitted the next morning.
  - Start VRII infusion at on the morning of the IOL, if BG is >7.0 mmol/L on two consecutive readings.

10.4 Caesarean Section

Instructions for caesarean section in patients with diabetes in pregnancy (where caesarean section is indicated)[3]:

- Patients on Metformin should omit the dose the night before and the morning dose and should not be started on VRII.
- Patients on basal-bolus insulin should:
  - Receive their basal insulin as normal the night before the caesarean.
  - Patients on morning basal insulin doses should receive half their regular dose on the morning of the induction. This is particularly important in patients with T1DM.
  - All other insulin doses should be omitted the next morning.
  - Start VRII on the morning of the surgery.
- Intra-operatively BG monitoring:
  - Measure BG at least once every 60 minutes, if epidural anaesthesia is used and every 30 minutes if general anaesthesia is used[3].

View Pathway

10.5 Pre-Term Labour and the Use of Steroids

In women with diabetes in pregnancy in pre-term labour[3]:

- Diabetes is not a contraindication for steroid use.
- In the absence of contraindications, steroids should be given up to 36 weeks gestation.

Steroid administration[3]:

- Women with insulin-treated diabetes who are given steroids, should be admitted to the antenatal unit.
- If the patient is nil-by-mouth:
  - Start on IV insulin and dextrose infusion, using the VRII protocol described in Table 10.2.
- Patients on insulin:
  - Pre-meal short acting insulin doses should be increased by 30%.
  - Pre-meal insulin doses should also be adjusted for any existing pre-meal hyperglycaemia (as shown in Table 10.5 below) **[R-GDG]**.
  - This may need to be continued for at least 24 hours after the last dose of corticosteroid.
- Patients with GDM, not on insulin:
  - Should be admitted for two hourly glucose monitoring if their SMBG results are >8 mmol/L.
  - If BG continues to rise, follow the subcutaneous sliding scale in (Table 10.5) and consult the diabetes team.

Blood glucose monitoring **[R-GDG]**:

- BG is likely to increase 9-15 hours after the first steroid injection.
- Glucocorticoids worsen postprandial hyperglycaemia. Therefore, up to 50% increase in pre-meal insulin dose may be required.
- If the BG level is >12 mmol/L, monitor the fetus by CTG until normoglycaemia is achieved.
- The insulin regimen should be closely monitored and adjusted by diabetes team during woman's admission.

Blood glucose (mmol/L)	Pre-Meal Supplemental Subcutaneous Insulin Sliding Scale Use only bolus insulin (i.e. aspart (Novorapid) or lispro (Humalog)	
	Patients with Insulin-Treated Diabetes	Patients with GDM on Diet-Control
< 4.0	Treat hypoglycaemia then give 20% of usual insulin dose	No insulin required
4.0 - 8.0	Usual dose	No insulin required
8.1 - 10.0	Usual dose + 2 units	3 units
10.1 - 12.0	Usual dose + 4 units	5 units
12.1 - 14.0	Usual dose + 6 units	7 units
14.1 - 16.0	Usual dose + 8 units	9 units
>16	Check ketones and start IV insulin infusion	Check ketones and start IV insulin infusion

Table 10.5: Pre-meal supplemental subcutaneous insulin sliding scale **[R-GDG]**.

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11 Postnatal Care

11.1 Neonatal hypoglycaemia

Neonatal blood sugar should be tested within one to two hours of birth for neonatal hypoglycaemia (< 2.6 mmol/L). For infants with blood glucose levels < 2.6 mmol/L, supplementary breastfeeding where possible is recommended [70].

11.2 Glycaemic Control

Women with pre-existing insulin-treated diabetes[3]:

- Following delivery, insulin requirements decline rapidly.
- For women with T1DM[3]:
  - A postpartum insulin regimen should be prescribed prior to delivery if possible.
  - This should commence immediately following the third stage of labour.
- For women with T2DM (insulin-treated)[3]:
  - The doses of insulin should be discussed with the diabetologist.

Women with T2DM diabetes, who were not treated with insulin prior to pregnancy[3]:

- If treated with insulin during pregnancy, women may be switched back to oral hypoglycaemic agents following delivery.

Women with GDM[3,8]:

- Insulin therapy should be discontinued immediately postpartum.
- SMBG should be discontinued, once BG levels return to normal.
- T2DM should be suspected and investigated, if hyperglycaemia persists
- Women with GDM have a 35–50% chance of recurrence in future pregnancies.
- All women should have 75g OGTT performed 4-12 weeks post-partum, patients who missed the OGTT within the 12 weeks check should have a HbA_{1C} performed **[R-GDG]**.
- Perform annual monitoring of HbA_{1C} or fasting BG.
- It is recommended for women who were diagnosed with prediabetes to follow intensive lifestyle interventions with or without Metformin to prevent diabetes[2].

11.3 Breastfeeding

Women with GDM should be encouraged to breastfeed their infants as soon as possible after delivery and at frequent intervals (every two to three hours) and to continue for at least three to four months postpartum to avoid neonatal hypoglycaemia and/or to prevent childhood obesity and diabetes in the offspring and to reduce risk of type 2 diabetes and hypertension in the mother[2,12,27,63,64,68].

Women should be advised[3,27]:

- There is an increased risk of hypoglycaemia while breastfeeding.
- Mothers may require less insulin due to the calories expended with breast feeding and may require a carbohydrate-containing snack before or during breastfeeding.
- Diabetes medications which were discontinued for safety reasons in the pre-conceptual or antenatal period should continue to be avoided during lactation.
- Women should maintain frequent contact with the diabetes service during the postpartum period to allow for glycaemic assessment and insulin dose adjustment.
- Metformin, glibenclamide and insulin are all considered compatible with breast feeding and if needed can be restarted **[R-GDG]**.
- Women who intend to formula-feed their infant may recommence diabetes therapy as per their pre-pregnancy management.

11.4 Postnatal Health Education and Promotion

Women with pre-existing diabetes:

- Health education should emphasise the importance of[2,3] **[L2]**:
  - Achieving and maintaining an ideal body weight.
  - Sustaining optimal glycaemic control.
- Appropriate lifestyle changes should be advocated, e.g.[2]:
  - Smoking cessation.
  - Healthy eating.
  - Physical activity.
- Pre-conceptual counselling for future pregnancies, should be discussed with the woman and her husband , with emphasis on the importance of tight glycaemic control and high dose folic acid supplementation in the pre-conception and early pregnancy period[3].
- Methods of contraception agreeable with the woman and her husband should be discussed and prescribed as appropriate[3].
- Referral to routine diabetes care should be made in the postnatal period **[R-GDG]**.

The mothers should be encouraged to stay in the hospital for at least 24 hours after birth **[R-GDG]**.

Babies of women with diabetes **[R-GDG]**:

- Should stay with their mothers unless there is a clinical complication or there are abnormal clinical signs that warrant admission for intensive or special care.
- Should be fed as soon as possible after birth (within 30 minutes) and then at frequent intervals (every two to three hours) until feeding maintains pre-feed capillary plasma glucose levels at a minimum of 2.0 mmol/litre.
- Should have their BG tested routinely at two to four hours after birth.

Consider admitting the babies of women with diabetes to the neonatal unit if they have **[R-GDG]**:

- Hypoglycaemia associated with abnormal clinical signs.
- Respiratory distress.
- Signs of cardiac decompensation from congenital heart disease or cardiomyopathy.
- Signs of neonatal encephalopathy.
- Signs of polycythaemia and are likely to need partial exchange transfusion.
- Need intravenous fluids.
- Need tube feeding (unless adequate support is available on the postnatal ward).
- Jaundice requiring intense phototherapy and frequent monitoring of bilirubinaemia.
- Been born before 34 weeks (or between 34 and 36 weeks if dictated clinically by the initial assessment of the baby and feeding on the labour ward).

Women with GDM:

- Education in the postnatal period should incorporate advice on[2,3]:
  - Diet.
  - Physical activity.
  - Weight reduction or healthy weight maintenance.
- The increased risk of developing T2DM in the future should be emphasised. Risk factors for developing T2DM include[2,3] **[L1]**:
  - Pre-pregnancy overweight/obesity.
  - High BG levels at diagnosis of GDM.
  - High insulin requirements during pregnancy.
  - Early gestation at diagnosis of GDM.
  - The need for insulin treatment in pregnancy.
  - Pre-term delivery.
  - Post-partum OGTT results in keeping with pre-diabetes.

- Future pregnancy planning should include evaluation of glycaemic control and if present, hyperglycaemia should be treated prior to conception[3].
- Women with identified postnatal glucose intolerance or confirmed diabetes should be advised to seek pre-conceptual counselling for any future pregnancies[2].
- Methods of contraception agreeable with the woman and her partner should be discussed and prescribed as appropriate[3].

Post Partum Glycaemic Screening in Women with GDM:

All women with a history of GDM should be offered one or more the following[2]:

- 75 grams oral glucose tolerance test between 4 – 12 weeks and up to six months post-delivery.
- FBG and HbA_{1C} three to six months post-delivery.
- In women with evidence of pre-diabetes or with NPRP risk category "high-risk gestational diabetes" in pregnancy should be screened annually thereafter.
- Women with a history of GDM should have lifelong screening for the development of type 2 diabetes or prediabetes every one to three years.

12 Key Considerations for Patient Preferences

Patient preferences refer to patient perspectives, beliefs, expectations, and goals for health and life, and to the steps employed by individuals in assessing the potential benefits, harms, costs, and limitations of the management options in relation to one another. Patients may have preferences when it comes to defining their problems, identifying the range of management options and selecting or ranking the outcomes used to compare these options.

It is important for healthcare professionals to develop an understanding of the patient as an individual and the unique way in which each person experiences a condition and its impact on their life.

The following recommendations are therefore made for physicians and other healthcare professionals regarding general principles of patient care in Qatar:

- **Respect Patients:** Treat patients with respect, kindness, dignity, courtesy and honesty regardless the patient dresses or appears. Ensure that the environment is conducive to discussion and that the patient's privacy is respected, particularly when discussing sensitive, personal issues. Ask the patient how they wish to be addressed and ensure that their choice is respected and used.
- **Maintain Confidentiality:** Respect the patient's right to confidentiality and avoid disclosing or sharing patients' information without their informed consent. In this context, students and anyone not directly involved in the delivery of care should first be introduced to the patient before starting consultations or meetings, and let the patient decide if they want them to stay.
- **Clarify Third-Party Involvement:** Clarify with the patient at the first point of contact whether and how they would like their partner, family members or carers to be involved in key decisions about their care or management and review this regularly. If the patient agrees, share information with their partner, family members or carers.
- **Obtain Informed Consent:** Obtain and document informed consent from patients, in accordance with MOPH policy and guidance.
- **Encourage Shared Decision Making:** Ensure that patients are involved in decision making about their own care, or their dependent's care, and that factors that could impact the patient's participation in their own consultation and care including physical or learning disabilities, sight, speech or hearing impairments and problems with understanding, reading or speaking English are addressed.
- **Disclose Medical Errors:** Disclose errors when they occur and show empathy to patients.
- **Ensure Effective Communication:** Explore ways to improve communication including using pictures, symbols or involving an interpreter or family members. Avoid using medical jargon. Use words the patient will understand and confirm understanding by asking questions.
- **Ensure Continuity of Care:** Provide clear and timely sharing of patient information between healthcare professionals, especially at the point of any transitions in care.

13 Performance Measures

A list of performance measures is given in the table below. Healthcare organisations are encouraged to monitor service performance using the indicator definitions below.

Number	Numerator	Denominator
DP01	Number in the denominator who undergo a 75g oral glucose tolerance test at 24-28 weeks of gestation.	Total number of pregnant women seen for antenatal care in the last 12 months ≥ 24 weeks gestation.
DP02	Number in the denominator who are prescribed 5 mg/day folic acid from at least 3 months before conception.	Total number of pregnant women diagnosed with diabetes who have been seen for antenatal care in the last 12 months.
DP03	Number in the denominator who have had a HbA_{1C} level measured at their booking appointment.	Total number of pregnant women in whom diabetes was diagnosed prior to the pregnancy, who were seen for antenatal care in the last 12 months.

Table 13.1: Performance Measures.

14 Appendix: Update to the Literature Search

The updated literature search included articles published between August 2020 and August 2023 that included three relevant clinical trials included in the update. The literature was searched as described in Figure A.1 below:

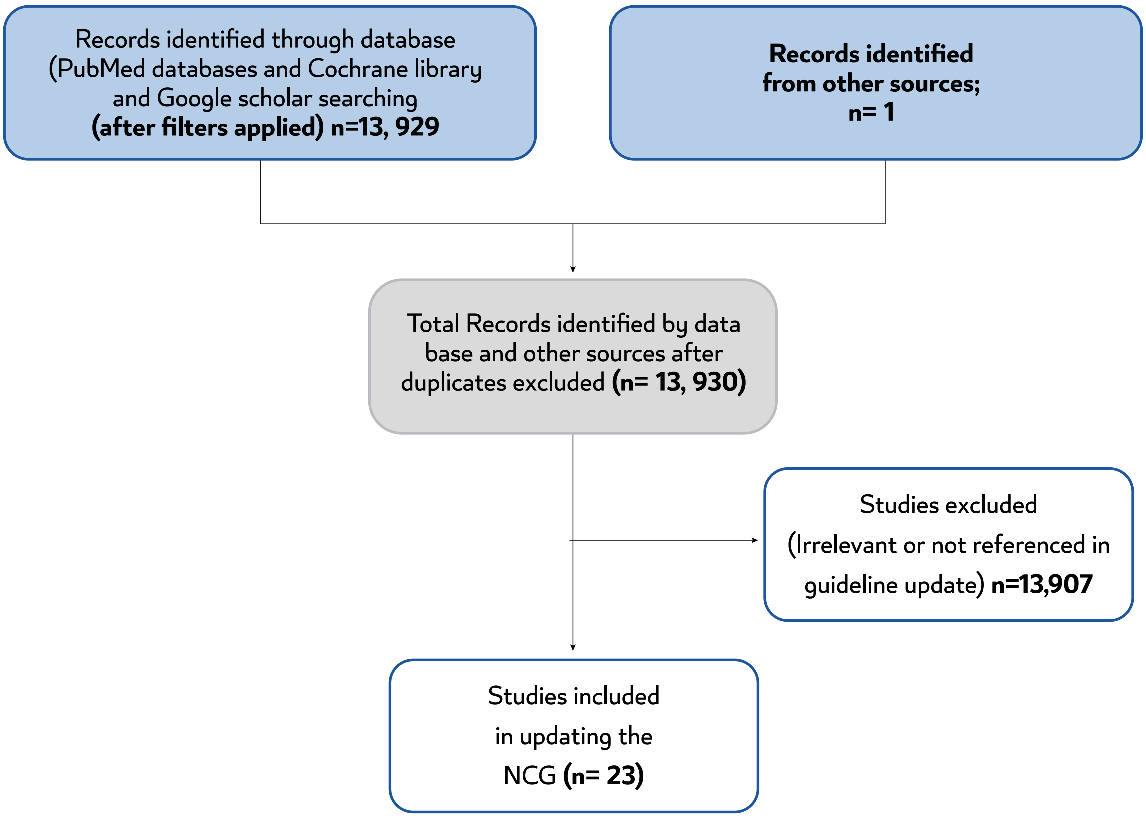

Fig A.1: Flow diagram for literature search results to update the NCG (2023).

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